

**MIDWIFERY OUTCOMES  
MANAGEMENT SYSTEM (MOMS)**

**Innovative Software Engineering Project  
Documentation**



**Higher Diploma in Software Engineering - 23.1F**

**National Institute of Business Management**

**GALLE**

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**Innovative Software Engineering Project**  
**Documentation**

**Submitted By**

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## **Declaration**

“We certify that this project does not incorporate without acknowledgement, any material previously submitted for a Diploma in any institution and to the best of our knowledge and belief, it does not contain any material previously published or written by another person or myself except where due reference is made in the text. We also hereby give consent for our project report, if accepted, to be made available for photocopying and for interlibrary loans, and for the title and summary to be made available to outside organizations.”

This project was submitted for the partial fulfilment of the Higher Diploma in Software Engineering course.

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## **Abstract**

Public Health Midwives provide community-level maternal and child healthcare and maintain a large volume of information concerning families, pregnancies, births, children, vaccinations, growth, nutrition and field activities. In a predominantly paper-based environment, the same information may be copied into several registers, reports require repeated manual calculations, and historical records are difficult to search or continue when staff members are transferred.

The Midwifery Outcomes Management System (MOMS) is a web-based information system designed to centralize these records and support the operational work of Public Health Midwives, supervisors and health administrators. The system provides modules for geographic service areas, staff, families, people, pregnancies, birth outcomes, children, growth measurements, developmental observations, vaccination, supplementation, clinics, appointments, field visits and report submission.

A relational database is used to preserve links throughout the maternal-and-child care lifecycle, while role-based access, validation and structured workflows improve consistency and accountability.

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## List of Acronyms and Abbreviations

- 1) **API** - Application Programming Interface
- 2) **BMI** - Body Mass Index
- 3) **DBMS** - Database Management System
- 4) **ERD** - Entity Relationship Diagram
- 5) **GRAVIDA** - Total number of pregnancies regardless of outcome
- 6) **HFAZ** - Height for Age Z-Score
- 7) **IDE** - Integrated Development Environment
- 8) **MCH** - Maternal and Child Health
- 9) **MOH** - Medical Officer of Health
- 10) **MOMS** - Midwifery Outcomes Management System
- 11) **MUAC** - Mid-upper Arm Circumference
- 12) **PARA** - Number of pregnancies that reached delivery regardless of outcome
- 13) **PDO** - PHP Data Objects
- 14) **PHM** - Public Health Midwife
- 15) **PHNS** - Public Health Nursing Sister
- 16) **RDBMS** - Relational Database Management System
- 17) **REST** - Representational State Transfer
- 18) **SQL** - Structured Query Language
- 19) **SPHM** - Supervising Public Health Midwife
- 20) **UI** - User Interface
- 21) **UML** - Unified Modeling Language
- 22) **WFAZ** - Weight for Age Z-Score
- 23) **WFHZ** - Weight for Height Z-Score
- 24) **WHO** - World Health Organization

## **Acknowledgement**

We would like to express our sincere thanks to my project supervisor, Mrs. Shamali A. Kurukularusuriya, for her guidance, support, and valuable advice throughout the development of this project. Her knowledge and encouragement helped us successfully complete this project.

We would also like to thank Ms. Thilini Dharmasena, the Course Coordinator, for her support and guidance during our Higher Diploma in Software Engineering program. We are grateful to all the lecturers of the National Institute of Business Management (NIBM) for sharing their knowledge and helping us improve our skills and understanding in software development.

We would also like to thank Mrs. N.T. Saumya, a Public Health Midwife in the Akmeemana area, and everyone who provided information and support for this project. Their ideas and feedback helped us understand the requirements of the system.

Finally, we would like to thank our parents, family members, and friends for their continuous support, encouragement, and motivation throughout the project. Without their help and support, completing this project would not have been possible. Thank you all for helping us complete the Public Health Management System successfully.

# **Chapter 1. Introduction**

## **1.1 Introduction of the Organization**

The Medical Officer of Health (MOH) office is an important government health institution in Sri Lanka. It provides many health services to people in the community. The MOH office works with Public Health Midwives (PHMs), Public Health Inspectors (PHIs), and other health staff to improve the health of mothers, children, and families.

Maternal and child health is one of the major responsibilities of the MOH. Public Health Midwives register pregnant mothers and children, record child growth, check development milestones, maintain vaccination records and provide health advice to parents. These services help children to develop healthily and support mothers during pregnancy and after childbirth.

Based on observations and discussions conducted at the requirement gathering phase, most of these records are still maintained with physical documents. As a result, finding relevant information, updating information, and preparing reports can take a lot of time. The other disadvantage of physical documentation is, physical documents can be lost or become unreadable with time.

The proposed Midwifery Outcomes Management System aims to solve these issues. It's a web-based system that makes it easier to manage health records. The system enables administrators, MOH officers, Public Health Midwives and parents to access and manage information. It has functions such as child registration, pregnancy registration, growth monitoring, observation milestone management, vaccination management and report generation.

## **1.2 Organization Structure**

The Medical Office of Health (MOH) office has an organized structure to provide healthcare services to people in the community. Different officers and staff work together to improve the health of mothers, children, and families. Each person has a different responsibility, and together they help provide a better health service.

### **Medical Officer of Health (MOH)**

The MOH is the head of the MOH office. The MOH manages all public health activities in the area. The officer supervises the staff, makes important decisions, and ensures that health services are provided properly.

**Public Health Nursing Sister (PHNS)**

The PHNS supervises the Public Health Midwives. She gives instructions, checks their work, and helps improve the quality of maternal and child healthcare services.

**Public Health Midwives (PHMs)**

Public Health Midwives work directly with families in the community. Their responsibilities include registering pregnant mothers and children, recording child growth measurements, checking whether the growing children follow development milestone standards, maintaining immunization records, and giving health advice to parents and community.

**Public Health Inspectors (PHIs)**

Public health inspectors help to keep the public health. They inspect the environment, assist in disease prevention and collaborate with the MOH office on community health programs.

**Administrative Staff**

The administrative staff carries out office tasks including record keeping, report preparation, document maintenance and support of daily MOH office activities.

**Parents and Guardians**

Parents, guardians are also intended to use the system to see relevant information about their family. They can register family details, view child health records, check vaccination information and get important health updates.

**1.3 Current Operations in the Organization****Maternity Care**

Public Health Midwives register pregnant women and visit them periodically. They take care of the health of the mother, give health advice and maintain pregnancy records, and birth outcome records.

**Child Registration**

When a baby is born, the child is registered to the MOH office. The child's personal information and family details are collected for future health care services.

**Growth Monitoring**

The height, weight and other growth details of the child are measured by the Public Health Midwives during the clinic visits. They are used to see whether the child grows normally or whether there are concerns regarding growth.

### **Development Milestone Monitoring**

Early childhood behaviors can be a strong indicator for mental growth assessment. At different ages, the physical and mental growth of the child is checked. If there is a delay, the child can be sent for further medical care.

### **Vaccination Services**

National immunization is another service provided by the MOH. The children are given vaccines as per the national immunization schedule. The details of the vaccinations are entered and checked at the clinic visits once the vaccines are administered.

### **Reports and Record Management**

The MOH office keeps health records and makes reports as needed. Currently, much of this information is still recorded on paper, so it takes longer to get the information and prepare reports.

### **Parent Services**

Parents can view their child's health records, vaccination dates and clinic appointments. They can also update their contact information and receive important health information from the MOH office.

## **1.4 Users and Responsibilities in the Proposed System**

The proposed Midwifery Outcomes Management System is designed to support the activities of Public Health Midwives and provide supervisory staff with access to relevant community-level information. The system has five main categories of users: Public Health Midwives, Supervising Public Health Midwives, Public Health Nursing Sisters, Medical Officer of Health, and parents. Each user group has different responsibilities and permissions according to their role in the system.

**Public Health Midwives (PHMs)** are the primary users of the proposed system and are responsible for entering and maintaining most of the community level information. They can register new families and family members, record pregnancy information, and maintain maternal and child health records. PHMs can record birth outcomes and register newborn children after delivery. They can also record information collected during field visits conducted for pregnant mothers and newborns. At health clinics, PHMs can record child growth measurements and monitor the growth and development of children. The system also supports PHMs in recording vaccinations and government-provided nutritional supplements

administered to children. In addition, PHMs can provide advice and guidance to parents based on the health and development information available in the system.

**Supervising Public Health Midwives (SPHMs)** have a supervisory role rather than performing routine data entry. Their main responsibility is monitoring the work carried out by PHMs. The system provides access to community-level statistics, summaries, and reports that can assist SPHMs in identifying areas requiring attention and evaluate whether the PHMs work is satisfactory.

**Public Health Nursing Sisters (PHNSs) and Medical Officer of Health (MOH)** have a higher-level management role within the system. They can access community level statistics and reports to monitor maternal and child health services within their responsible areas.

**Parents** are also intended as a user group of the proposed system. The system is intended to provide parents with access to selected information related to their family, and children.

## **1.5 Problem Definition**

The management of maternal and child health information in an MOH area involves maintaining records for a large number of families, pregnant mothers, and children. In the existing system, much of this information is recorded and maintained using registers, books, forms, and other paper documents. PHMs have to record information collected during field visits, clinics, and other health related activities in these records.

Managing a large number of paper records creates several practical difficulties. Records may become damaged through regular handling or may be lost when they are moved between different locations. Locating information about a particular mother, child, or family can also take considerable time when the required information is stored among a large number of physical records. This can make it difficult for PHMs to obtain previous information when providing follow-up services.

Maintaining up-to-date information is another challenge. Maternal and child health records are continuously updated as new events occur. For example, a child's growth measurements, vaccination history, development observations, or supplementation records may change over time. When these details are maintained manually, information has to be entered into the relevant records each time. Manual recording can result in incomplete entries, inconsistent information, or errors during data entry.

Supervisory staff such as SPHMs, PHNSs, and MOHs require summarized information to monitor the maternal and child health services provided within their areas. With paper-based records, obtaining an overall view of community-level information is difficult because the required data is distributed across individual records maintained by PHMs. This limits the ability to quickly review information and identify areas that may require attention.

Therefore, the main problem addressed by the proposed MOMS System is the difficulty of maintaining, retrieving, updating, and reporting maternal and child health information using paper documents. MOMS will be able provide a centralized digital system for managing family, pregnancy, birth, child, growth, vaccination, supplementation, and field-visit information. MOMS is intended to reduce unnecessary paperwork, make records easier to retrieve and update, support more accurate reporting, and provide PHMs and supervisory staff with information appropriate to their responsibilities.

## **1.6 Project Objectives**

The main purpose of the MOMS is to develop a web-based system that helps the MOH office manage maternal and child health records more efficiently. The system is designed to reduce paperwork, improve data management, and provide better healthcare services. The project aims to achieve the following objectives.

- 1. Improve Record Management**

Store maternal and child health records in a digital system instead of paper files. This makes it easier to add, update, and search records.

- 2. Reduce Paperwork**

Minimize the use of paper records and reduce the time spent on manual documentation and file management.

- 3. Improve Data Accuracy**

Reduce errors that happen during manual record keeping by allowing users to enter and update information directly in the system.

- 4. Support Parents**

Allow parents to view their child's health information, vaccination history, clinic dates, and other important records.

- 5. Generate Reports**

Help MOH officers and Public Health Midwives generate reports quickly and easily for monitoring maternal and child health services.

**6. Improve Security**

Protect health records by allowing only authorized users to access the system through secure login accounts.

**7. Save Time and Improve Efficiency**

Reduce the time required to manage records and prepare reports, allowing healthcare staff to provide better services to the community.

## **1.7 Proposed Solution**

The proposed solution is the Midwifery Outcomes Management System (MOMS), a web-based application developed to support the management of maternal and child health information at the MOH level. The system is intended to provide a central place for maintaining information that is currently handled through different paper-based records and forms.

The main users of the system are Public Health Midwives, who carry out most of the day-to-day activities supported by the application. PHMs can register families and their members, record pregnancy information, enter birth outcomes, register newborn children, and maintain records collected during field visits. The system also supports the recording of child growth measurements, and Thripasha supplementation.

SPHMs, PHNSs, and MOHs use the system mainly for supervision and review rather than routine data entry. They can view community-level information and reports that can assist them in monitoring the work carried out by PHMs and reviewing the health information recorded within their areas.

## **Chapter 2. Methodology**

### **2.1 Data Collection Methods**

Information for the development of the proposed system was collected by studying how maternal and child health information is handled in the field. The main focus was to understand the work carried out by PHMs, the records they maintain, and the information that needs to be available when providing services to families, mothers, and children.

Three methods were mainly used to gather the required information: interviews with PHMs, observation of their work, and analysis of PHM training documentation.

#### **1. Interviews with Public Health Midwives**

Interviews were conducted with Public Health Midwives to gain an understanding of their day-to-day activities and the way they maintain maternal and child health records. Questions were asked about the information they collect from families, the records maintained for pregnant mothers and children, and the activities carried out during field visits and clinics.

The discussions also helped identify difficulties associated with maintaining and finding information in paper records. Information obtained from the PHMs was used to understand which activities should be supported by the proposed system and what information needs to be recorded for each activity.

#### **2. Observation**

The work carried out by PHMs at the MOH office, and various clinics were observed to gain a better understanding of the actual process rather than relying only on information obtained through interviews. Attention was given to how records are maintained and how information is handled during different activities.

The observation included activities such as family and member registration, pregnancy-related record keeping, child registration, growth monitoring, recording development information, vaccination and supplementation records, and the preparation or use of reports. This helped to identify the sequence of activities and the points where a digital system could reduce manual work.

#### **3. Analysis of PHM Training Documentation**

PHM training materials and related documentation were reviewed to understand the standard procedures and records used in maternal and child health services. The documentation included

information about the forms and paper records used by PHMs and the way different types of information are expected to be recorded and maintained.

These materials were very useful when identifying the data fields required for the proposed system. They were also used to understand the relationship between different records, such as family, pregnancy, birth, child, growth, vaccination, and supplementation information.

Additionally, the child health record book provided by the World Health Organization (WHO) was studied to get a clear idea about WHO recommended growth standards.

The information obtained from the interviews, observations, and documentation analysis was compared and used during the requirements analysis and system design stages. This provided the base for deciding the main functions, data requirements, and workflow of the proposed MOMS system.

## **2.2 Software Development Tools**

The following are the set of development tools that were used for the implementation of MOMS System. They helped to design, develop, test, and maintain the system easier as the source code grows.

### **1. Visual Studio Code (VS Code)**

Visual Studio Code is a very capable and easy to use IDE with lot of extensions available to improve its functionality. It was used as the main code editor for developing the system. The importance of using a good IDE is it provide syntax highlighting, debugging, and extension support, which makes coding easier and faster.

### **2. HTML**

HTML is the default language of web. It is used to define web pages. Pure HTML without frameworks or libraries, was used to design all pages and other page elements.

### **3. CSS, Tailwind**

CSS (Cascading Style Sheets) was used to design the user interface. It helped create an attractive, responsive, and user-friendly layout for the system. Tailwind CSS library was used to add styles without having to manage large centralized style sheets.

### **4. JavaScript**

JavaScript was used to add interactive features to the system. It was used for form validation, search functions, dynamic content, and improving the user experience.

## **5. Chart JS**

Chart JS is a versatile and easy to use, HTML and JavaScript based library that can be used for data representation with dynamic charts and plots.

## **6. PHP**

PHP was used for server-side programming. It handled user requests, processed form data, connected with the database, and managed the system functions.

## **7. MySQL**

MySQL was used as the database management system. It stored information such as user accounts, child records, pregnancy records, growth measurements, development milestones, vaccination records, and etc. in relational database architecture.

## **8. Laragon**

Laragon was used as the local development environment. It provided the Apache web server and MySQL database, allowing the system to run and be tested on a local computer during development.

## **9. TablePlus, DBeaver**

TablePlus and DBeaver database management tools, in conjunction were used to create and manage the MySQL database. It made it easier to create tables, insert data, update records, and manage the database through a visual interface rather than depending on the text only console and queries.

## **2.3 Software Process Model**

The incremental Software Development Model was selected for developing the Public Health Management System. Since this project was developed by a small team, the incremental model was suitable because it allowed the system to be developed step by step. Each module was completed, tested, and debugged before moving to the next module.

## **2.4 Implementation Plan**

The implementation of the MOMS was carried out in several stages. These stages focused on developing the database, developing the full stack web application, implementing main functional modules, integrating the z-score calculation API service, continuous testing and debugging.

| Development Stage | Description                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage 1           | <b>Requirements Analysis, Environment and Project Setup</b><br>Identifying functional requirements, user roles, activities and responsibilities.<br>Selecting and configuring development tools, and software environment. |
| Stage 2           | <b>Database Development</b><br>Designing the MySQL database using ER notation and implementing the required tables, relationships, and constraints.                                                                        |
| Stage 3           | <b>Full Stack Web Development</b><br>Designing PHP based web application using a layered structure. (View, Controller, Service and Repository Structure)                                                                   |
| Stage 4           | <b>WHO Anthro API Integration</b><br>Developing a separate R based REST API to calculate child growth z-scores, using the WHO Anthro and Plumber packages.                                                                 |
| Stage 5           | <b>Testing and Debugging</b><br>Performing manual tests on implemented functionalities. Identifying errors and debugging.                                                                                                  |
| Stage 6           | <b>Documentation</b><br>Preparing and reviewing project report, database documentation, and UML documentation.                                                                                                             |

*Table 01: System Implementation Plan*

### **Database Development**

The MySQL database was created following ER notation to store user information, pregnancy records, child registration details, growth measurements, development milestones, vaccination records, and reports and etc.

### **Full stack Development**

PHP was used as the primary server-side programming language for the full-stack web application. The codebase was structured into View, Controller, Service, and Repository layers. It handles user login, data processing, database connections, and all system functions. HTML, CSS, and JavaScript were used to develop the user interface.

## **Chapter 3. Analysis**

This chapter analyses the existing system and the proposed Midwifery Outcomes Management System using Unified Modelling Language (UML) diagrams. UML is a tool that analyses, describe, and document software systems without leaning on programming heavy concepts.

The proposed system was analyzed and documented using the below UML diagrams to ensure that it matches with expected functional requirements.

The use case diagrams document the interactions between the user and the system without describing internal processes of the system. The class diagram visualizes the object-oriented structure of the proposed system to guide the software implementation for the developers. The sequence diagrams are used to describe and visualize internal processes and how each internal component interact with each other.

### **3.1 Use Case Diagrams of the System**

Figure 1 to Figure 4 visualize the user interactions with the system. The system provides services for four types of actors whom are parents, public health midwife, supervising public health midwife, and public health nursing sister. The parent actor almost only has information viewing ability relevant to them. The PHM is the main actor with most interactions with the system. The SPHM, and PHNS mostly have supervising capabilities within the system.

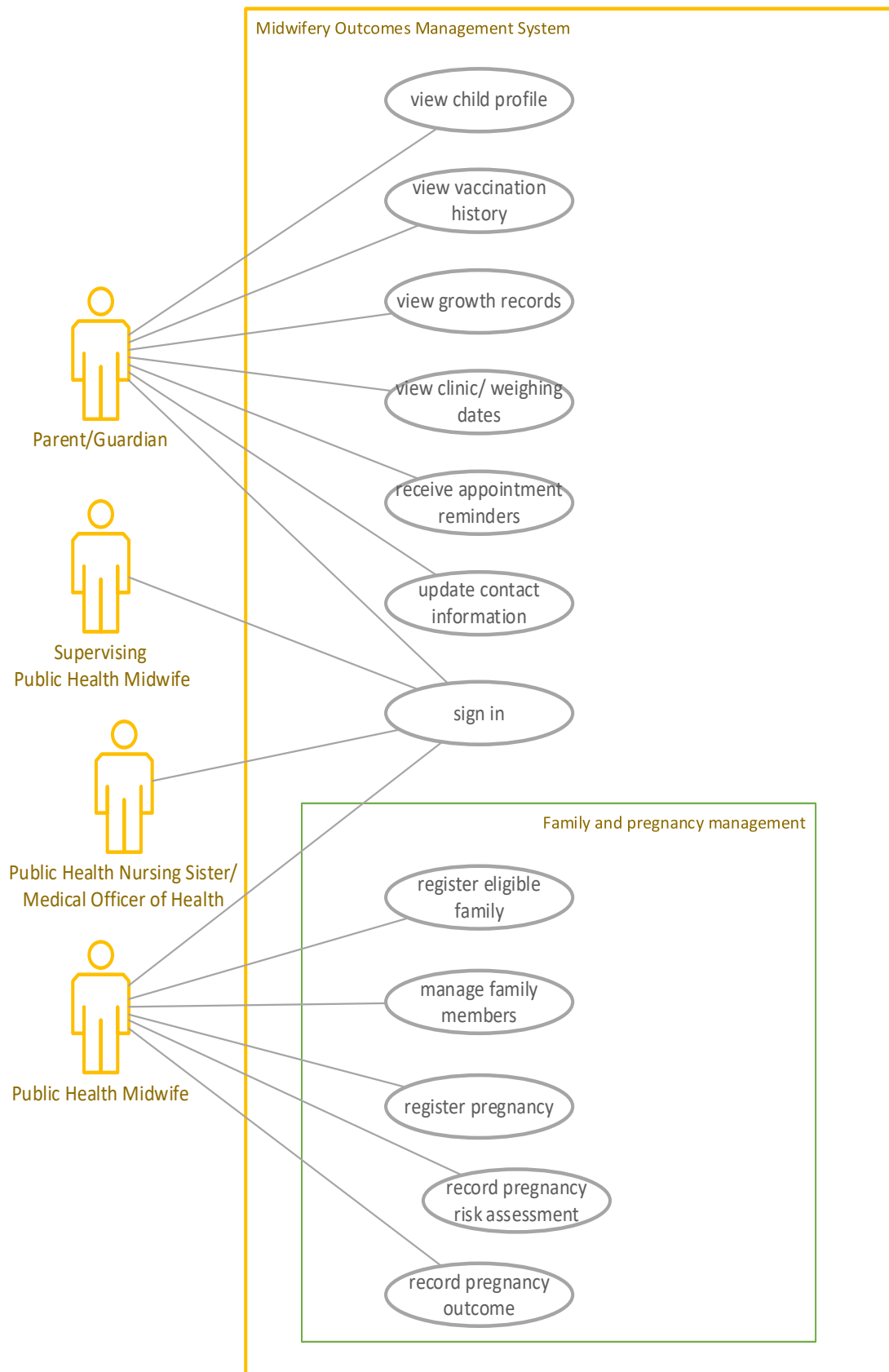


Figure 1: Use Case Diagram - part 1

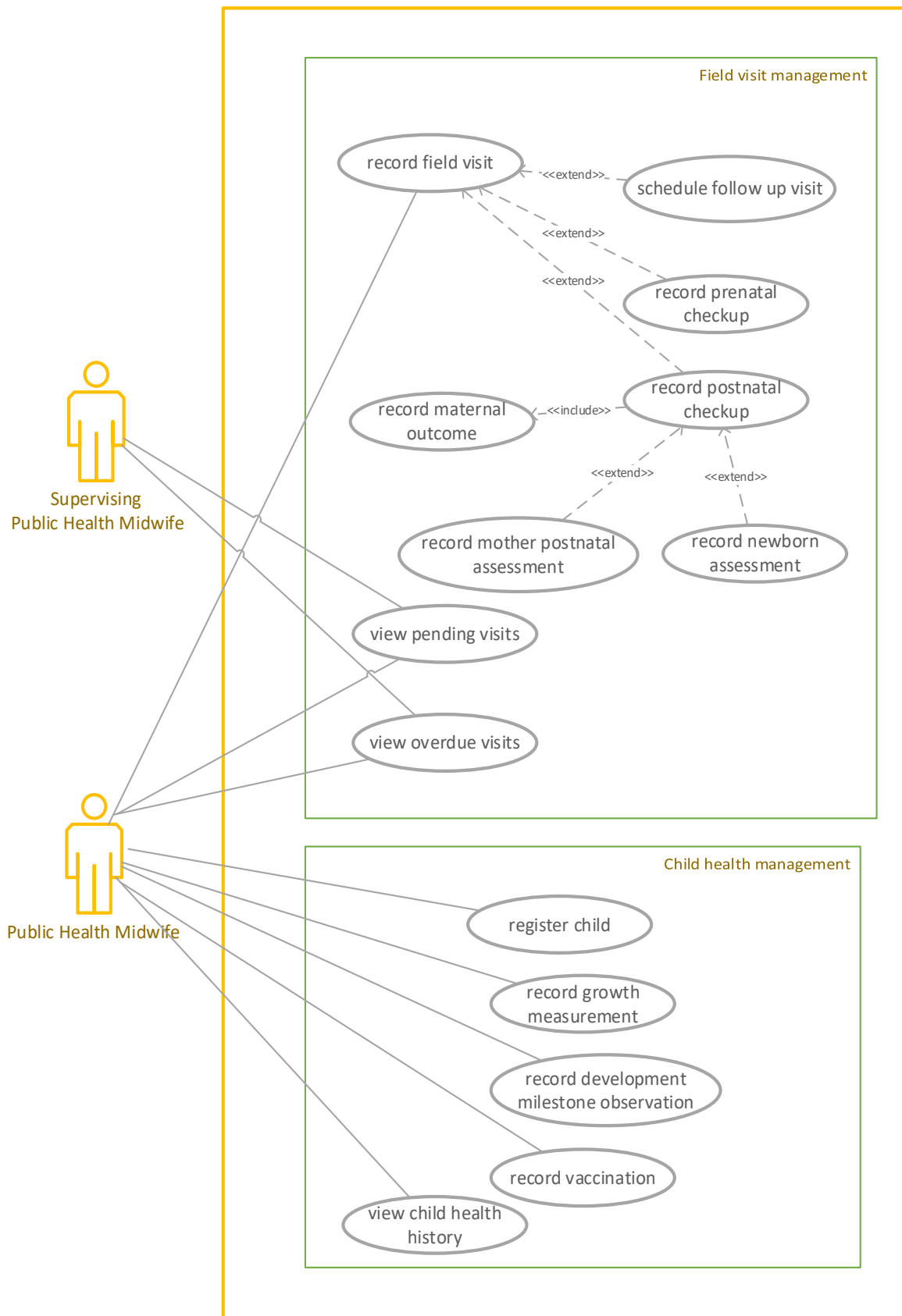


Figure 2: Use Case Diagram- part 2

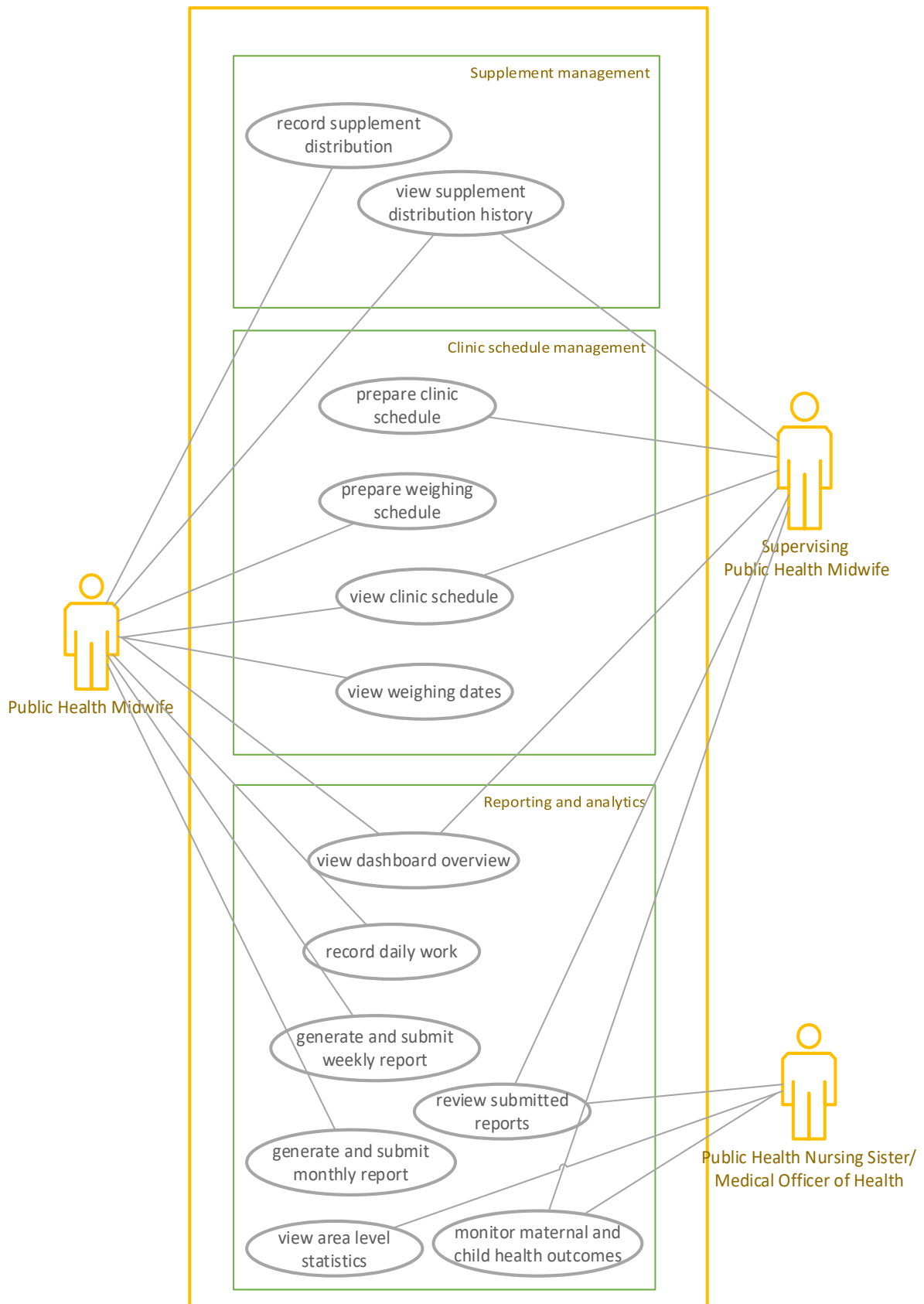


Figure 3: Use Case Diagram - part 3

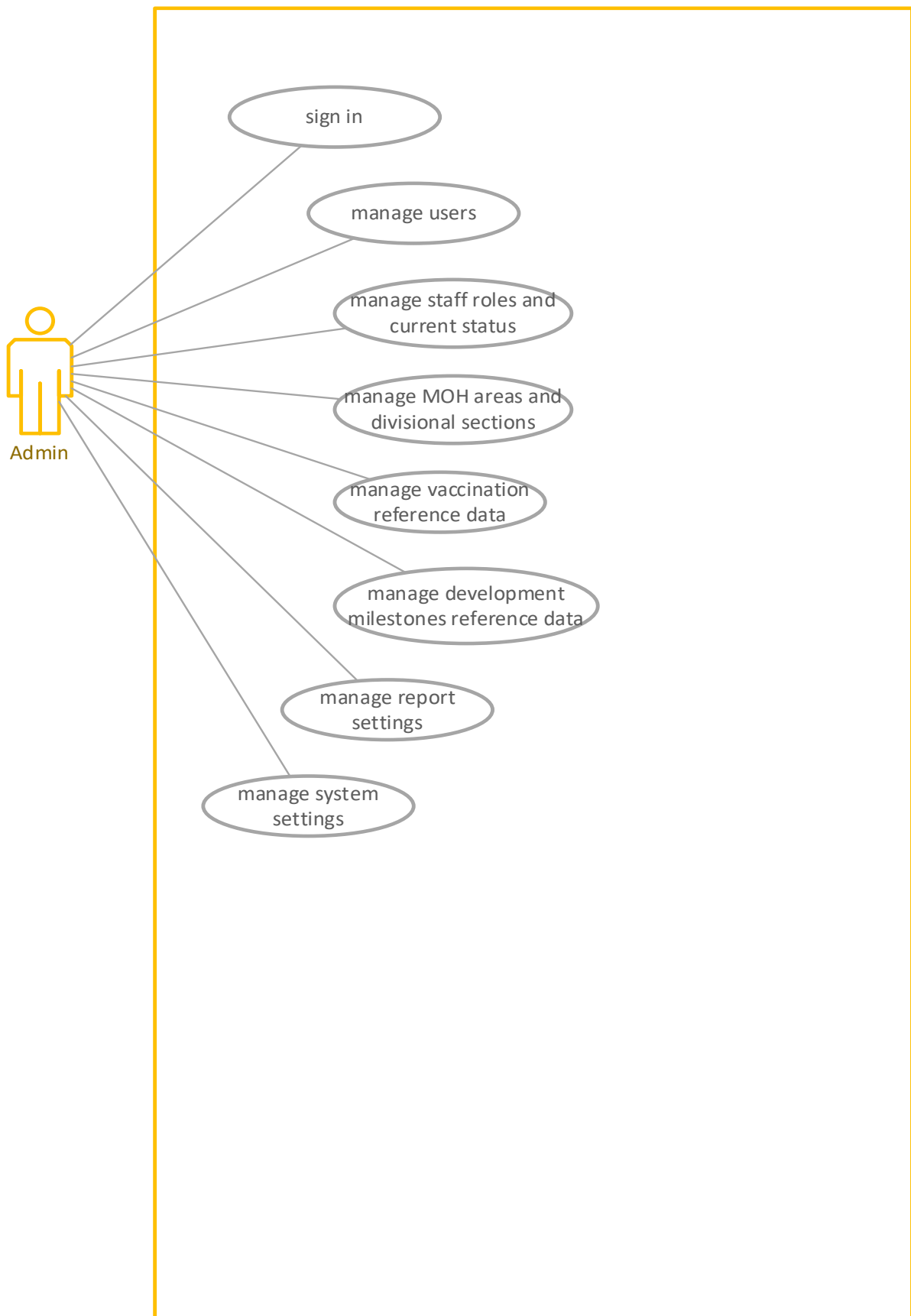


Figure 4: Use Case Diagram - part 4

## 3.2 Class Diagrams

The figure 5 and onwards shows the class structure of the proposed MOMS system. The system is designed and implemented with a multi-layer architecture with clear separation between layers of classes with specific functionalities. These layers are Controller classes: used to accept and respond to incoming requests, Service classes: that handles the business logic, and Repository classes: that are responsible for interactions with relational database. The controller level classes, and repository level classes inherit common behaviors from base controller class and repository class.

### 3.2.1 Users and Authentication Module

This diagram in figure 5 shows the class structure for user authentication management and user management.

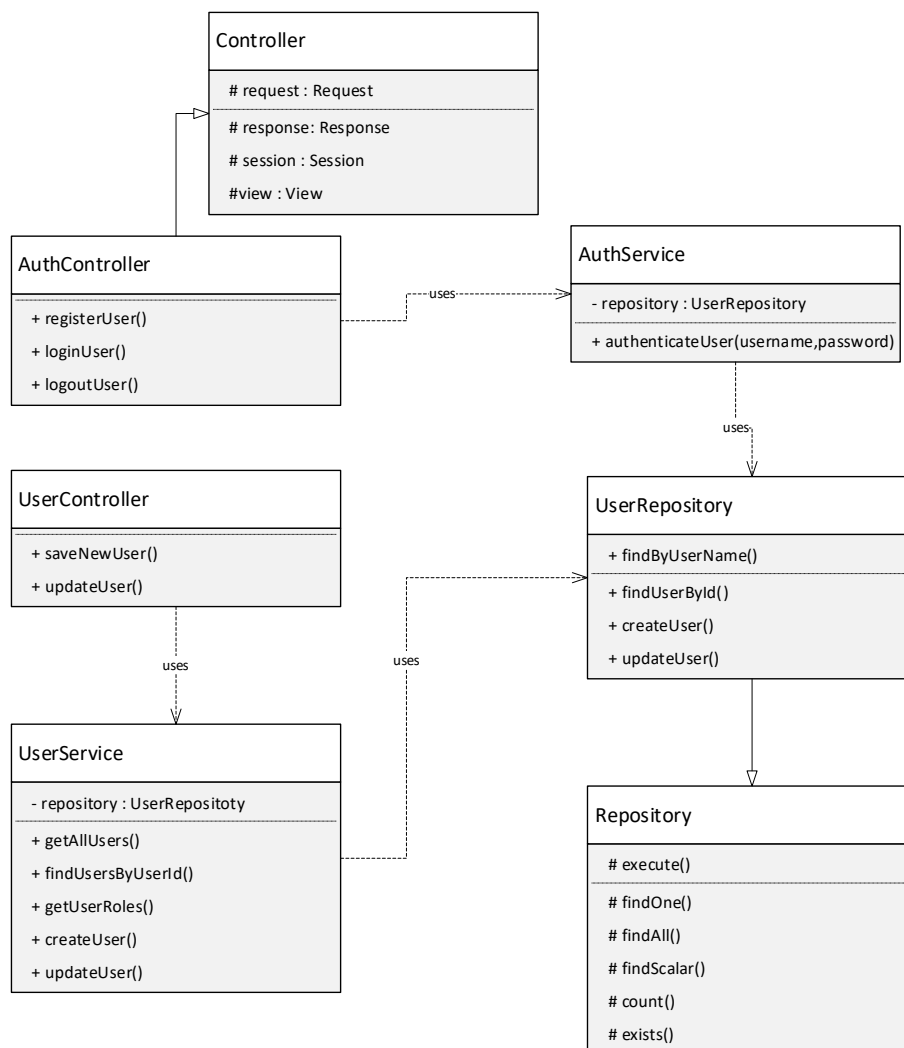


Figure 5: Class Diagram (User and Auth Module)

### 3.2.2 Dashboard Module

This diagram shows the class structure related to displaying a dashboard that summarize operational data in to one interface.

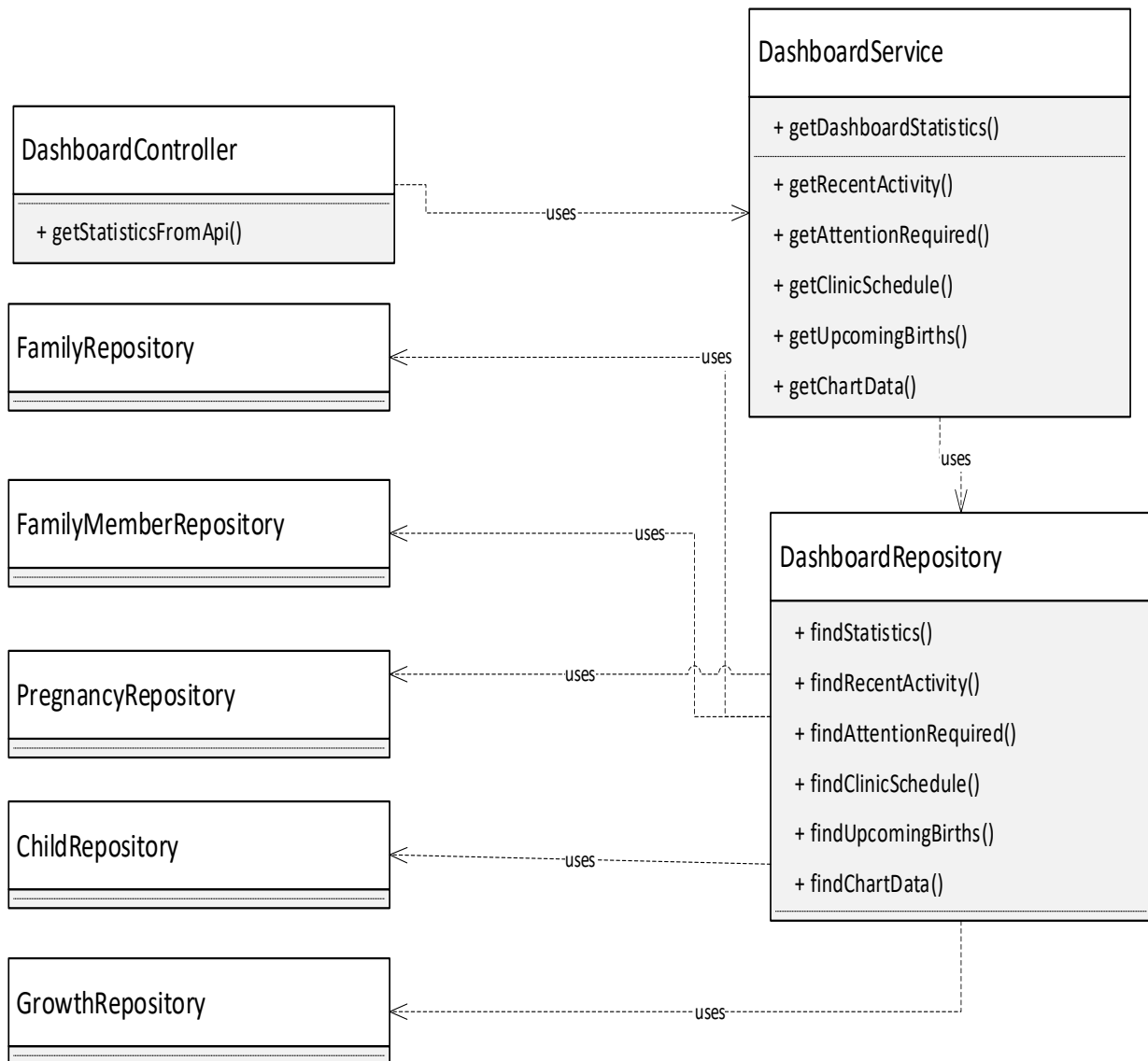


Figure 6: Class Diagram (Dashboard Module)

### 3.2.3 Family Module

This diagram shows the classes and associations that are responsible for parents and children family management module in the proposed system.

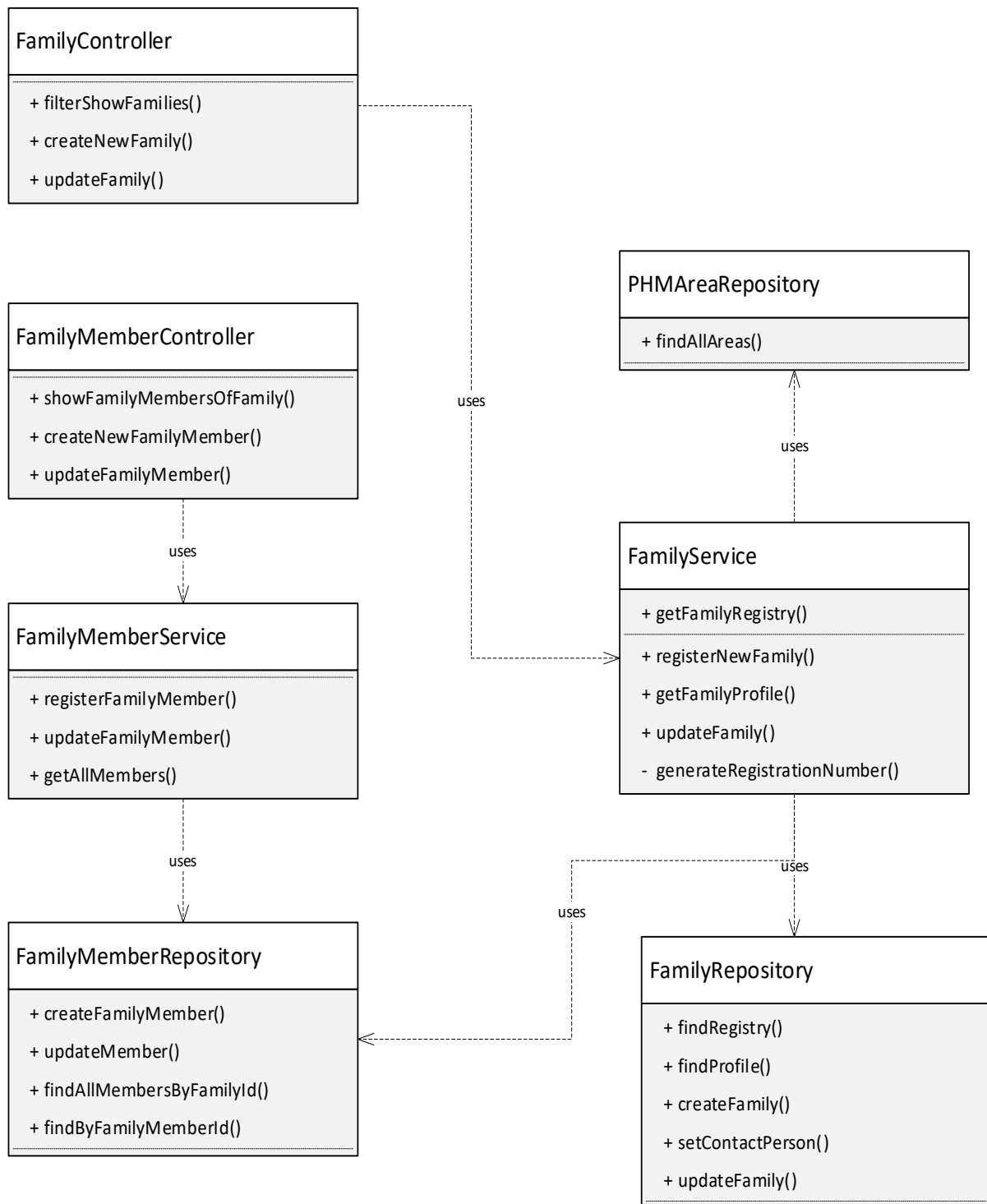


Figure 7: Class Diagram (Family Module)

### 3.2.4 Mothers Module

Shows the class structure responsible for retrieving information related to the mother module in the proposed system.

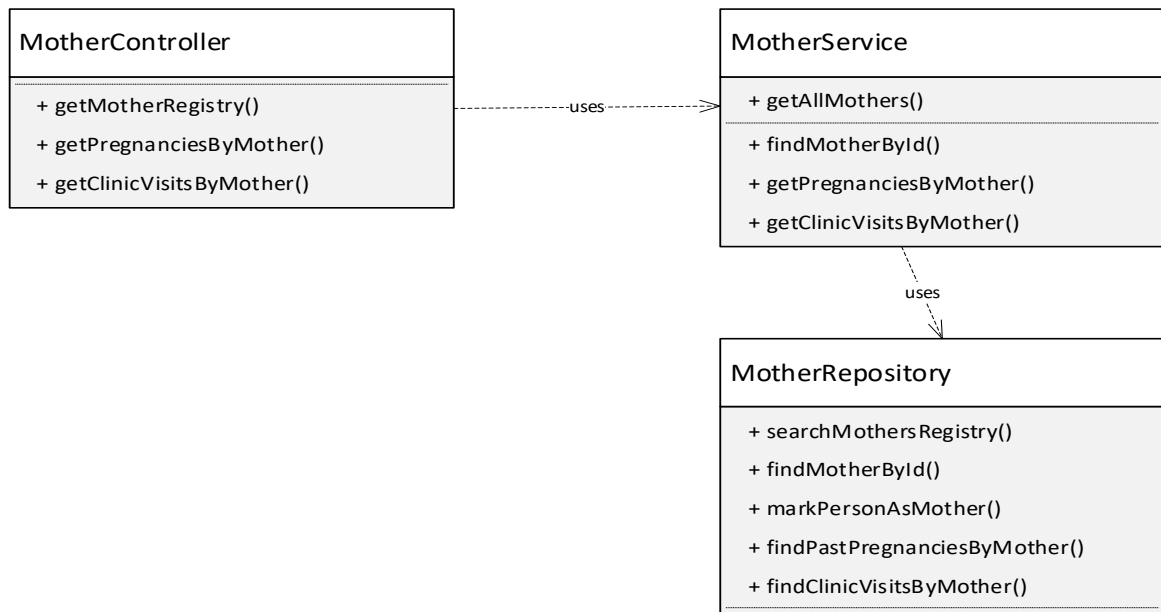


Figure 8: Class Diagram (Mothers Module)

### 3.2.5 Pregnancy Module

This class diagram shows the class structure that registers new pregnancies, update pregnancy status, and retrieve information requested by the user.

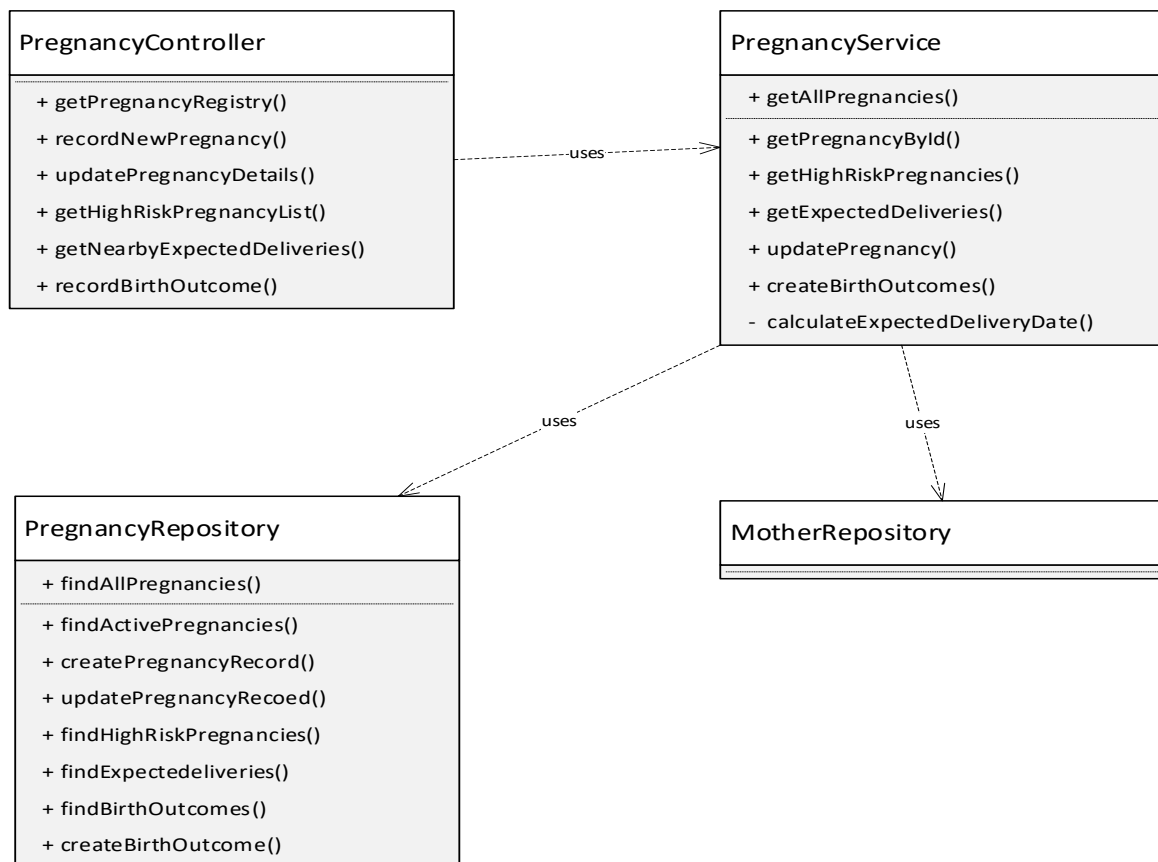


Figure 9: Class Diagram (Pregnancy Module)

### 3.2.6 Children Module

This shows the class structure responsible for the children module in the proposed system.

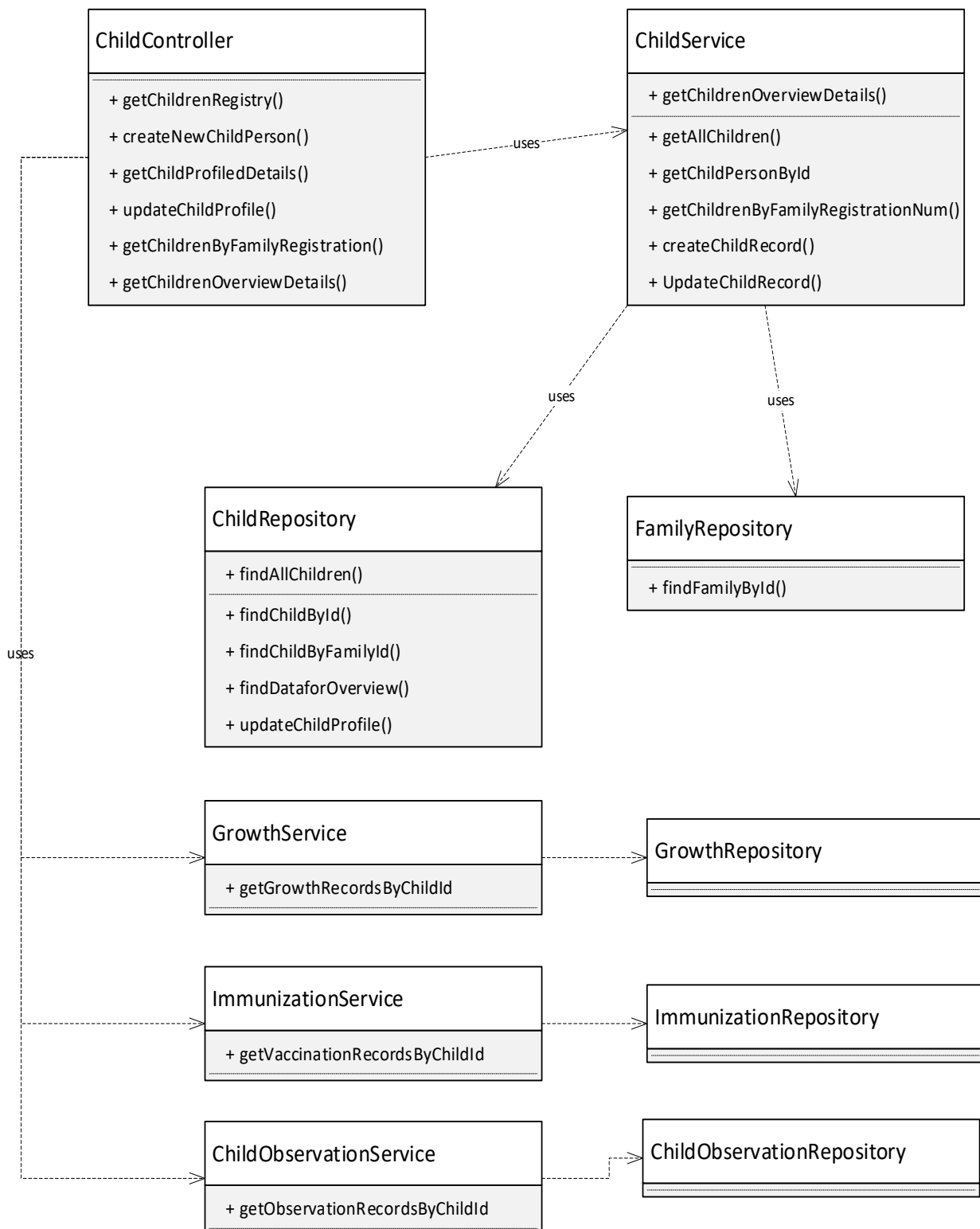


Figure 10: Class Diagram (Children Module)

### 3.2.7 Growth Module

Figure 11 shows the classes that records growth measurements for individual children, process them and save in the database.

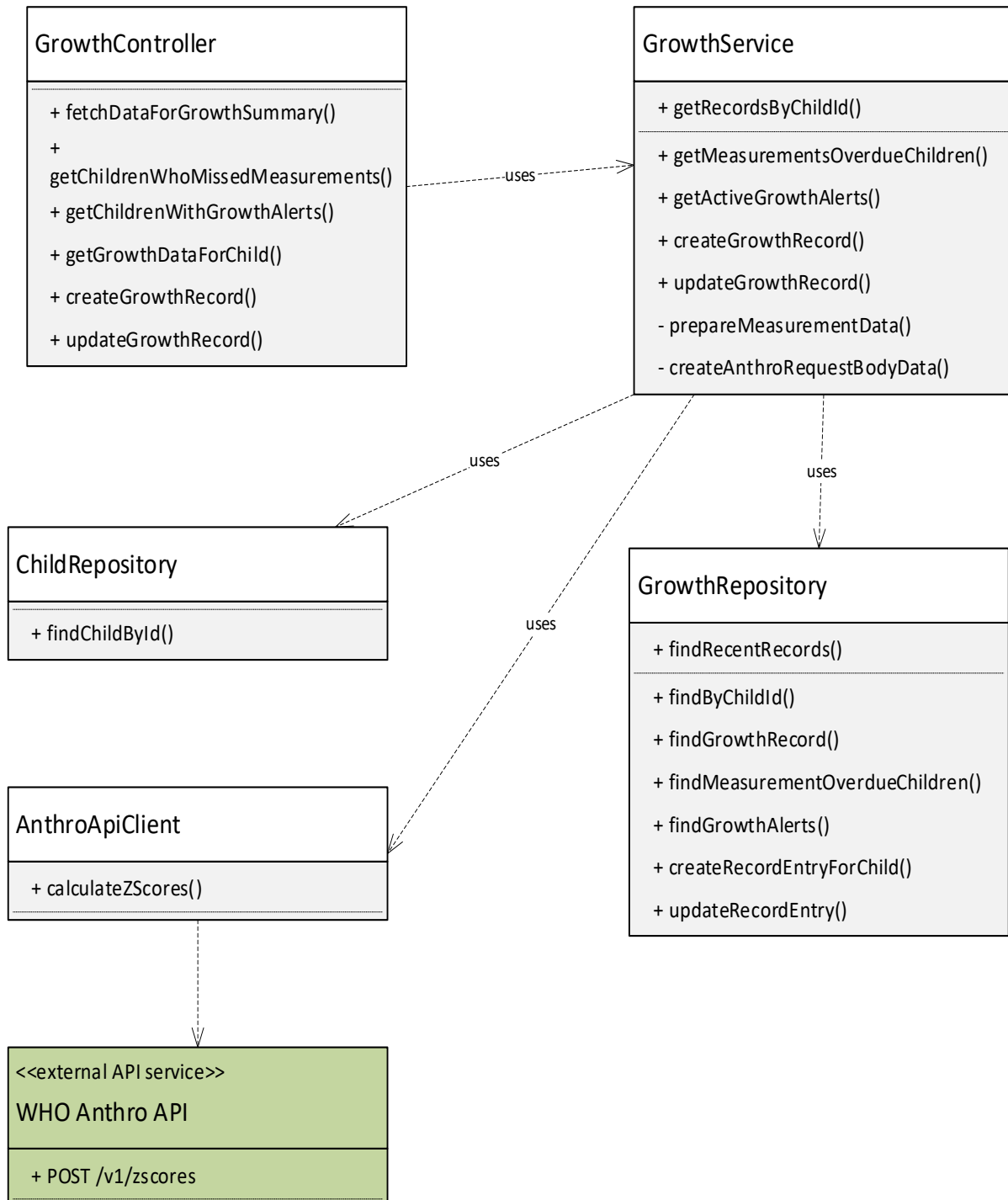


Figure 11: Class Diagram (Growth Module)

### 3.2.8 Developmental Observations Module

This shows the class structure responsible for recording, processing and archiving child development milestone related data.

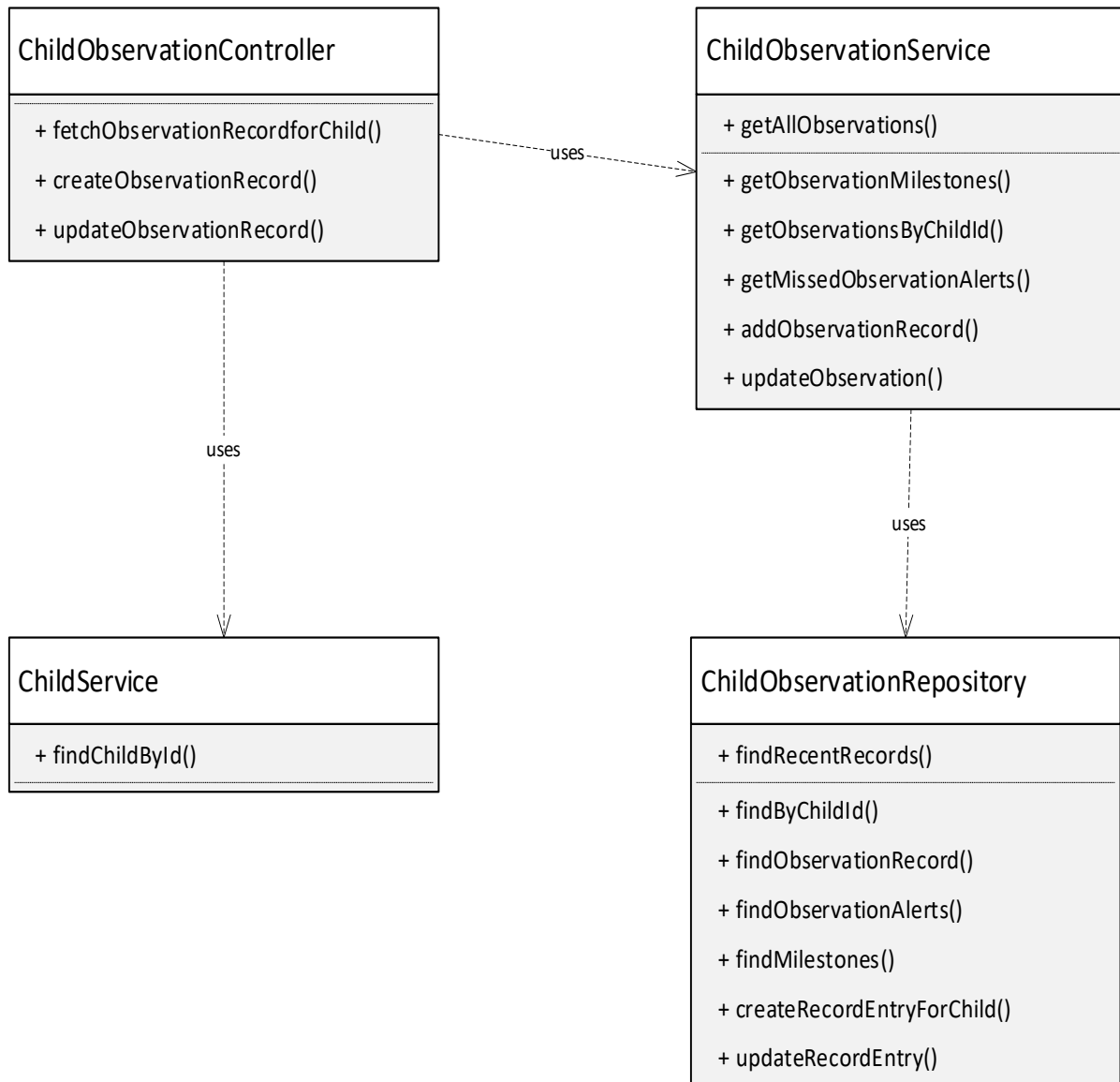


Figure 12: Class Diagram (Developmental Observations Module)

### 3.3 Sequence Diagrams

Figure 13 and following figures shows the interactions and data flow between components of the system. Initially the PHM provides user credentials to login to the system. Then the PHM can interact with the system to register new families, register family members, register pregnancies and pregnancy outcomes, register newborn children and their growth measurements. The system uses an external API service WHO Anthro API to calculate z scores for growth measurements.

#### 3.3.1 Log in

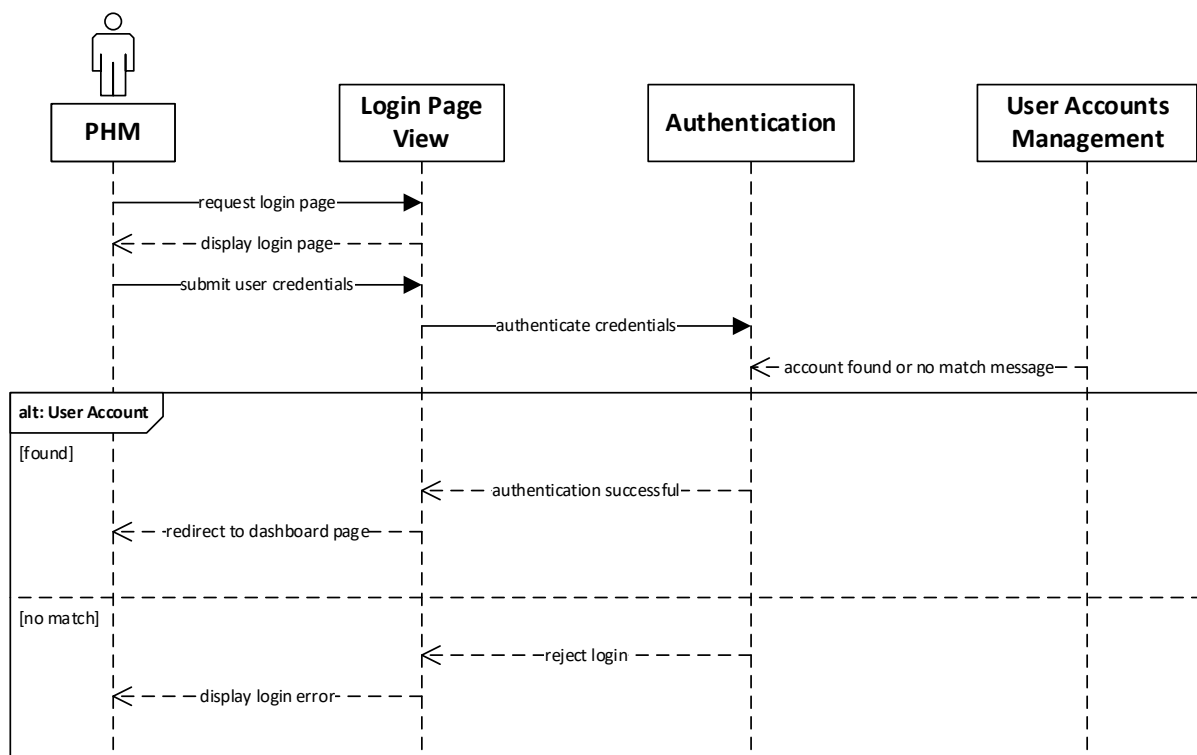


Figure 13: Sequence Diagram (Log In)

### 3.3.2 Register Family and Primary Contact

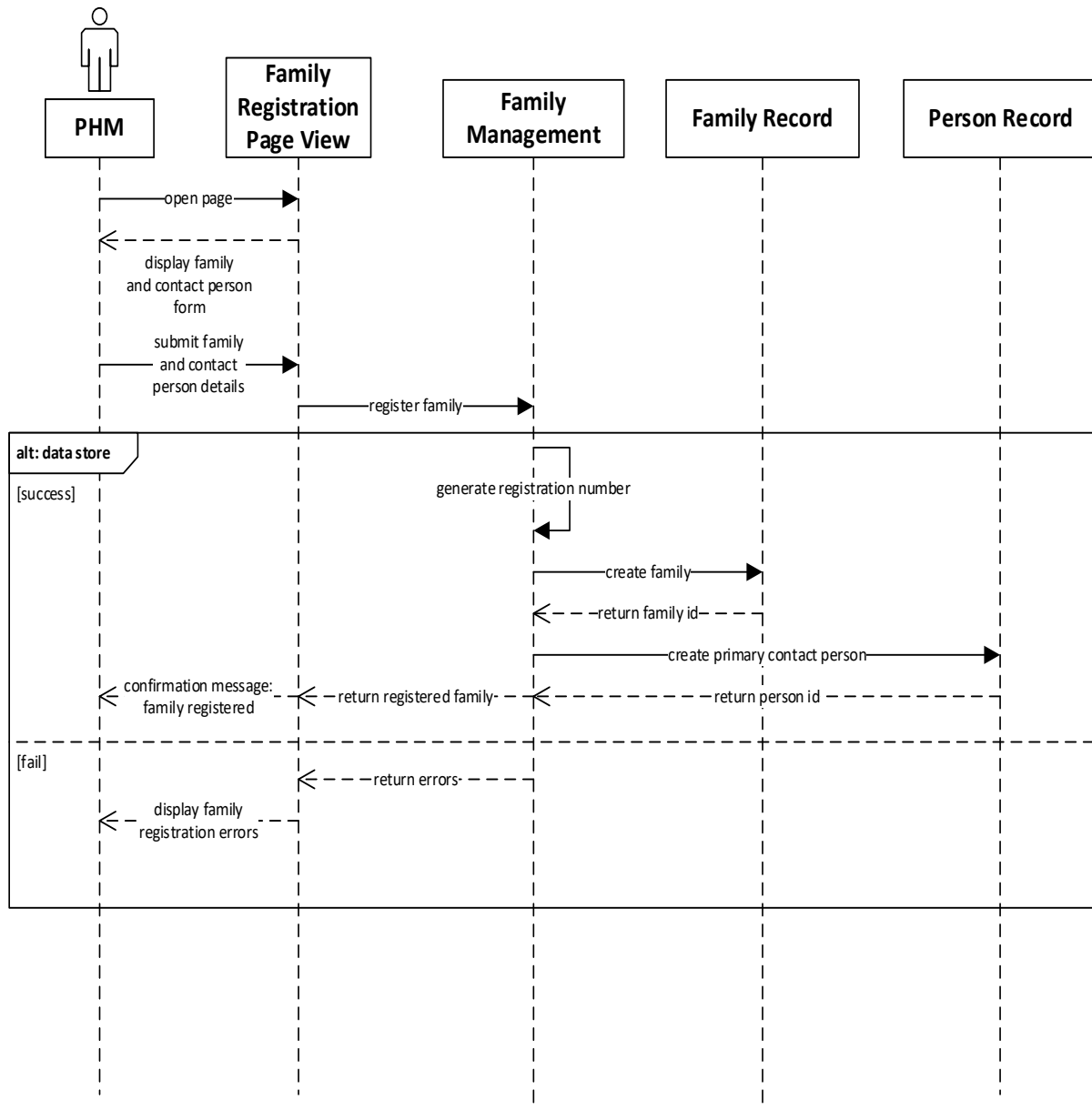


Figure 14: Sequence Diagram (Register Family and Primary Contact)

### 3.3.3 Register Family Member

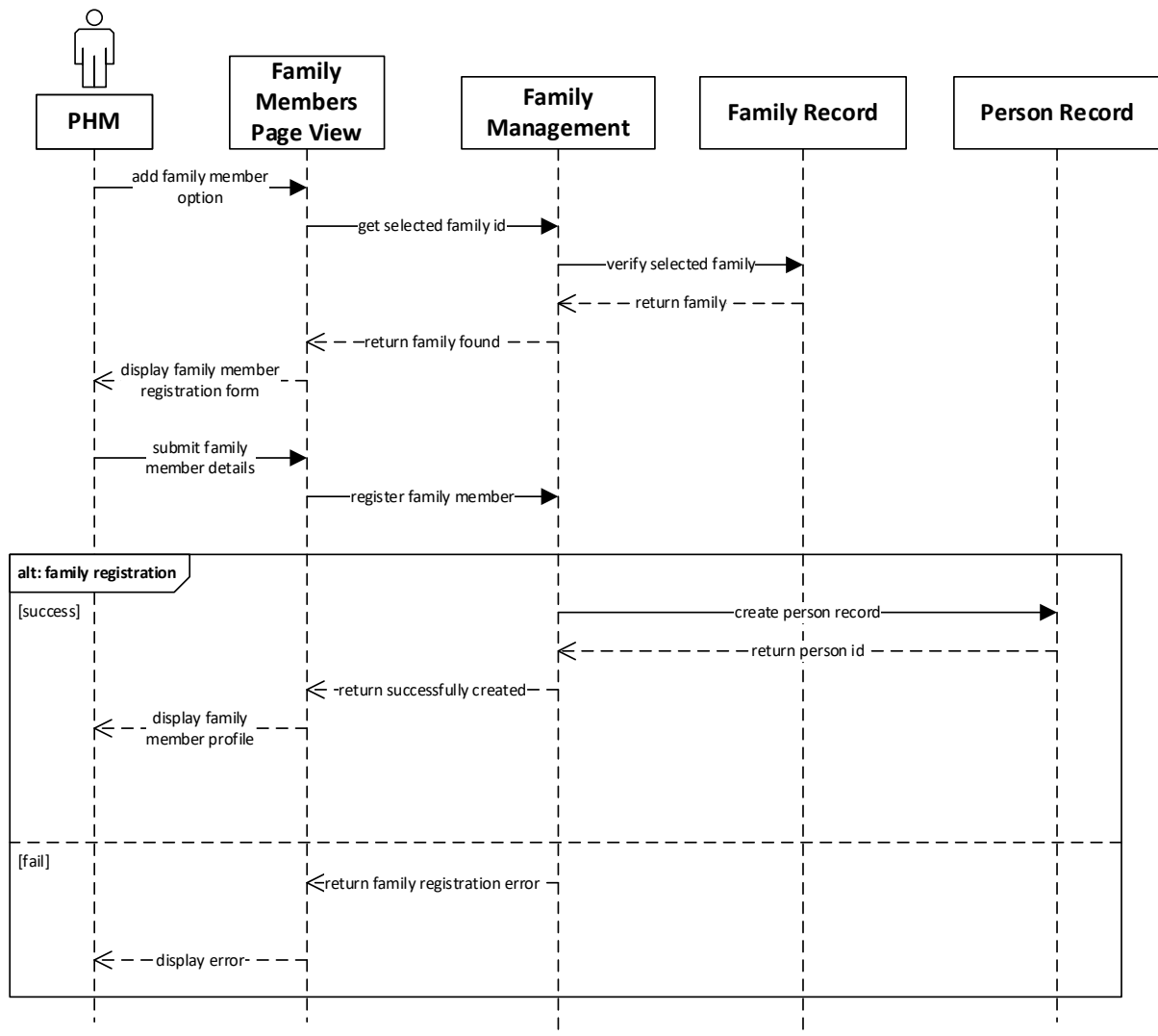


Figure 15: Sequence Diagram (Register Family Member)

### 3.3.4 Register Pregnancy

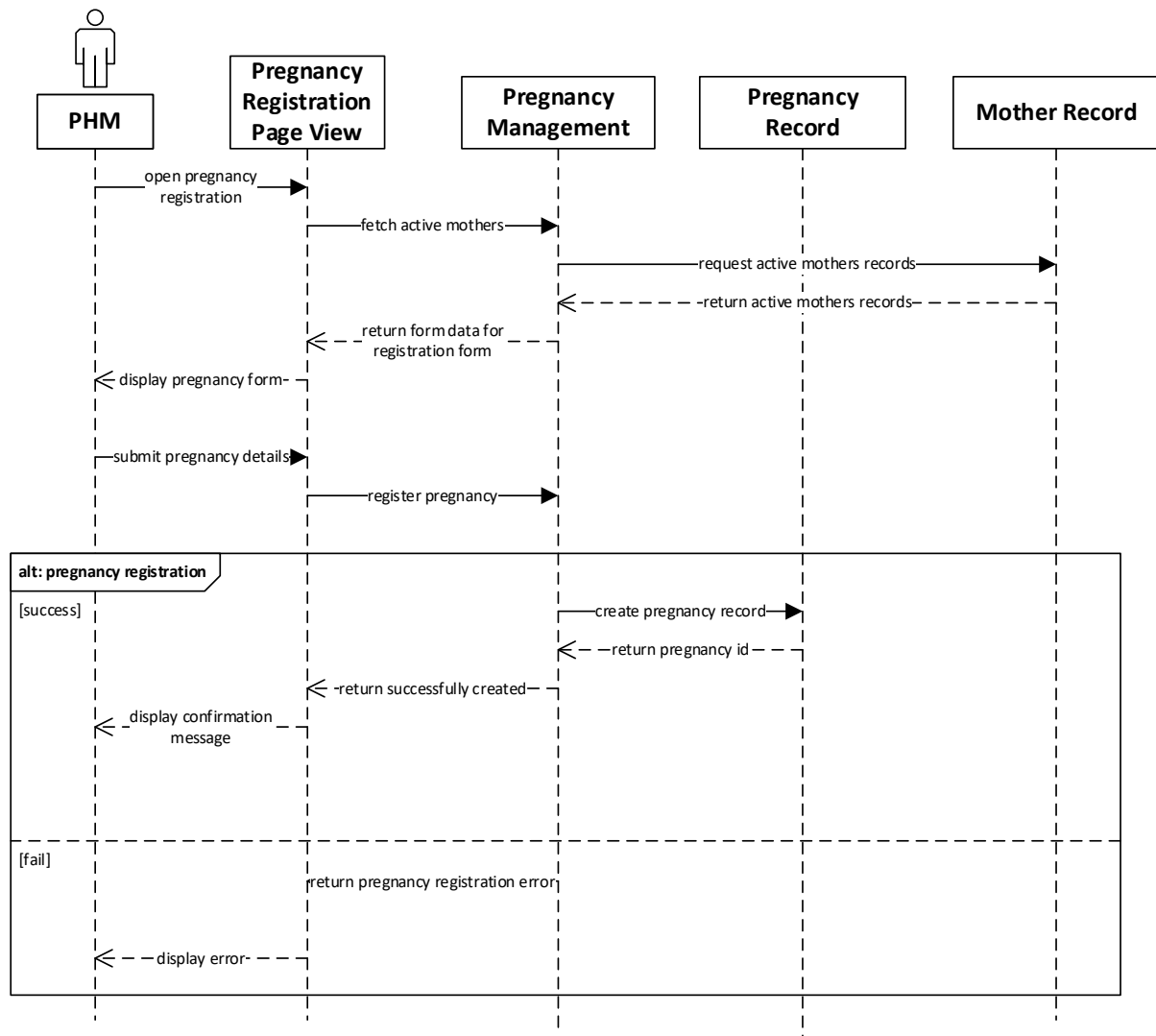


Figure 16: Sequence Diagram (Register Pregnancy)

### 3.3.5 Register Child from Pregnancy Outcome

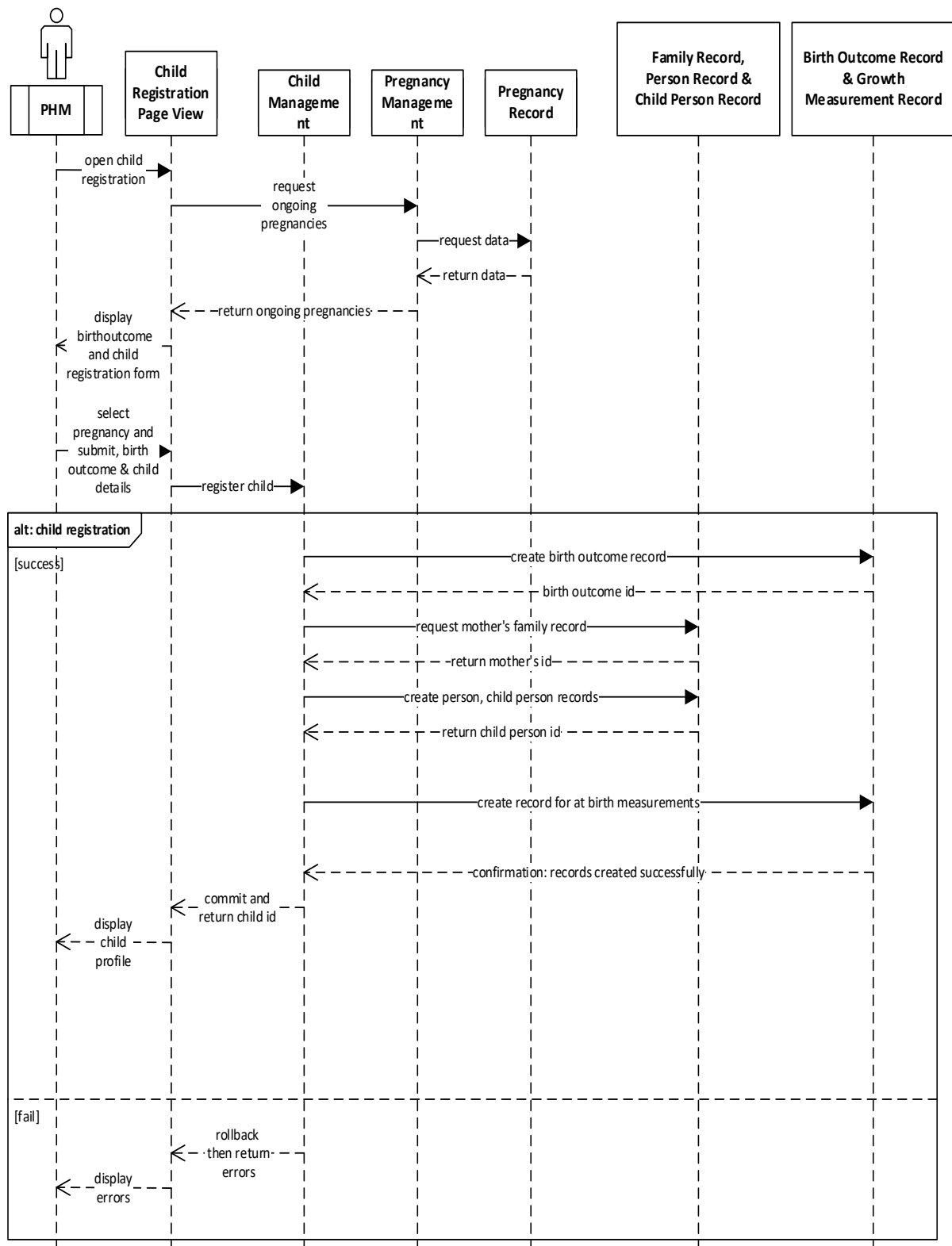


Figure 17: Sequence Diagram (Register Child from Pregnancy Outcome)

### 3.3.6 Record Growth Measurement

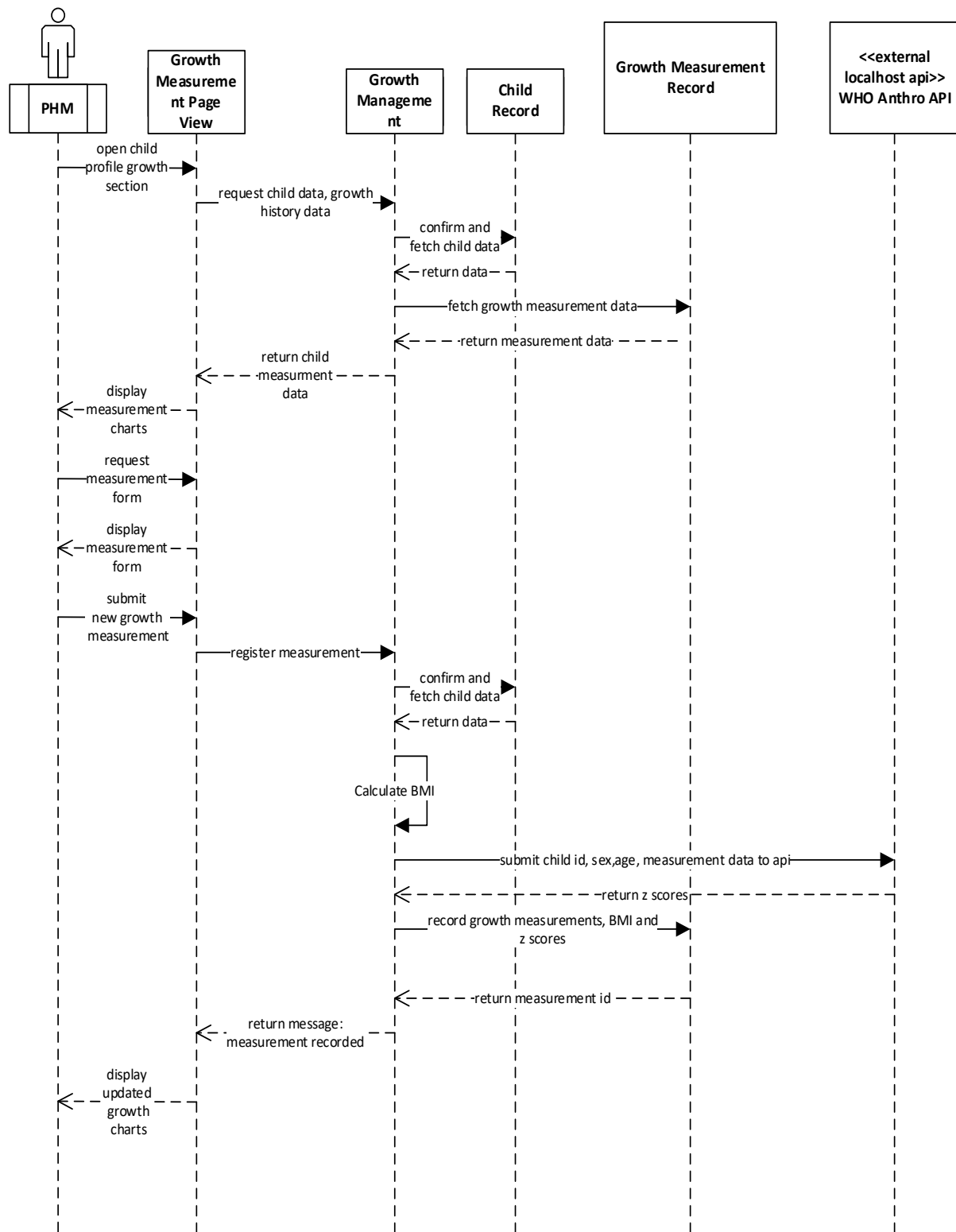


Figure 18: Sequence Diagram (Record Growth Measurement)

## Chapter 4. Solution Design

### 4.1 Architectural Design

Figure 19 shows the overall system architecture of the design that Chapter 5 implements. MOMS is a full-stack web application that follows client-server architecture. It consists of a web server application, a relational database, and a separate R-based (a programming language for statistical computing) REST API to calculate child growth z-scores via the Anthro package (also implemented in R) provided by the WHO.

The web application manages the main business logic and communicates with the relational database through the PDO interface. When child growth z-scores are required, the web application sends the measurement data to the above REST API through HTTP requests. The API, implemented using the Plumber package, uses the WHO Anthro package to calculate the growth-related z-scores. Then Plumber uses the jsonlite package to parse the results into JSON format and returns the results back to the web application.

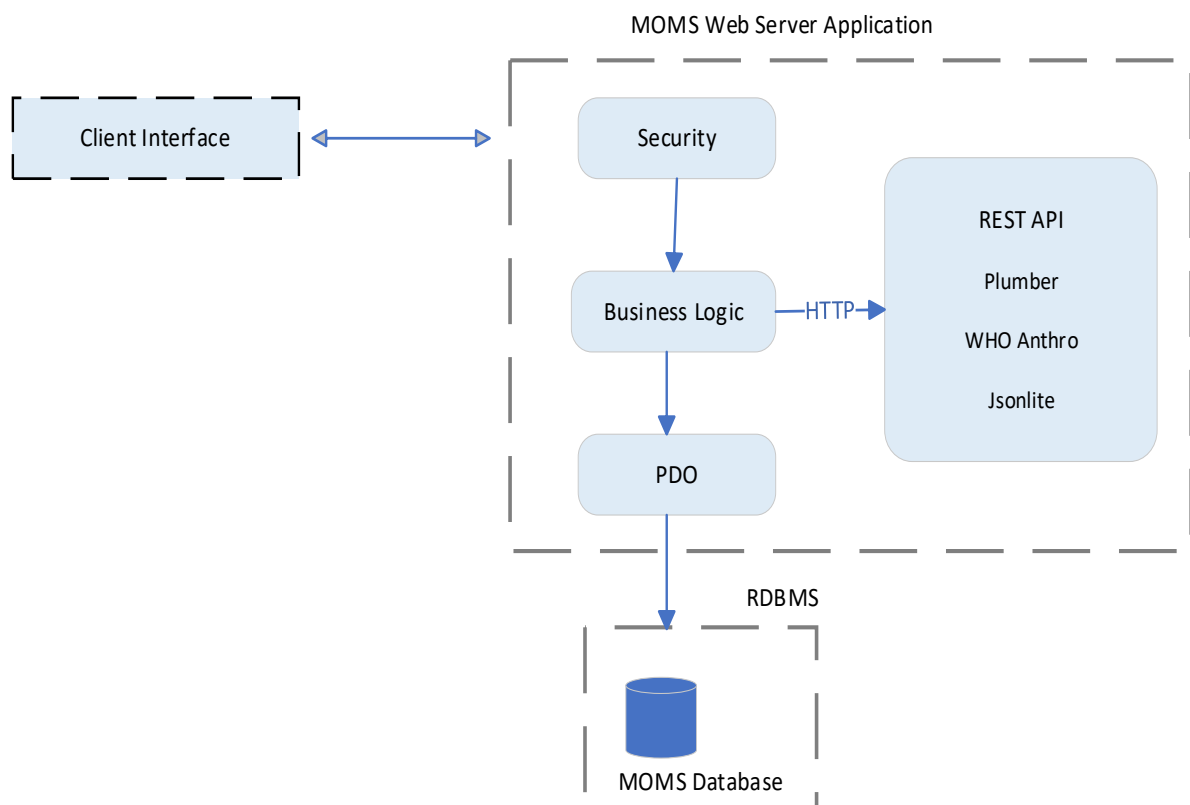


Figure 19: Architectural Diagram

## 4.2 User Interface Design

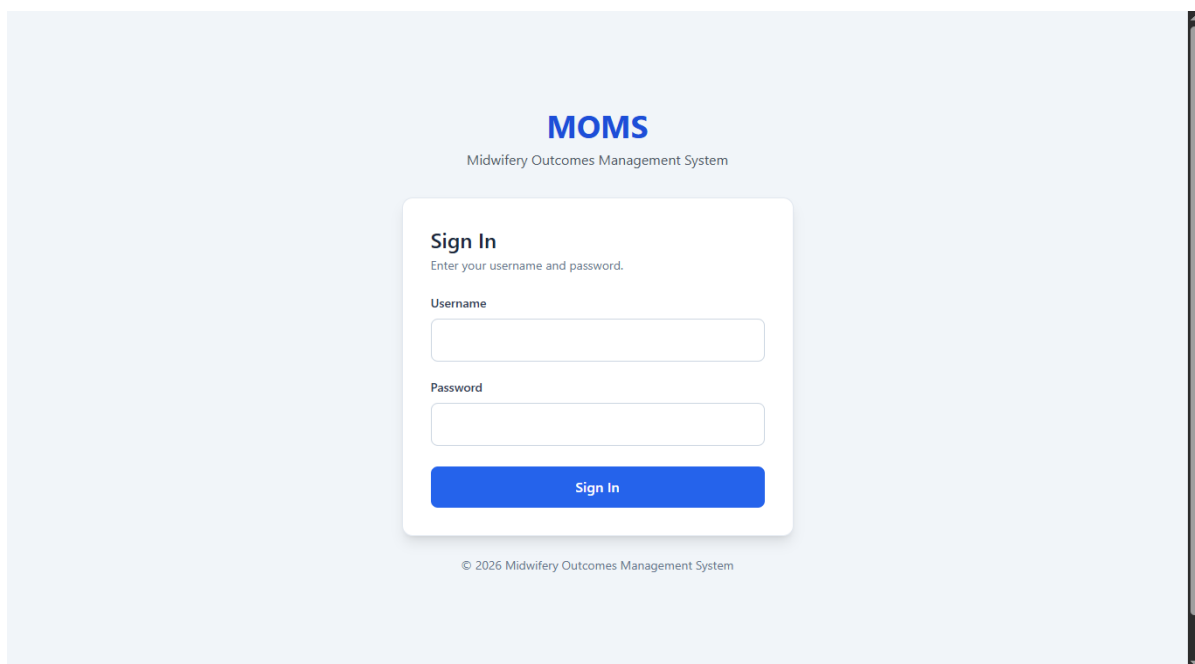
The main focus of the Midwifery Outcomes Management System interface design was to create a simple, clear, and user-friendly system. The design uses a clean layout with suitable colors and easy navigation to help users access functions quickly. All interfaces share a common design theme to keep the styling consistent.

Different interface layouts were designed for administrators, Public Health Midwives, and parents. Each user interface was created according to their responsibilities and system requirements.

The system pages were developed using HTML, Tailwind CSS, and JavaScript to provide a responsive and organized user experience. Forms, tables, and dashboards were designed to display healthcare information clearly and make daily tasks easier for users.

### Screen: 1

User Login Page - displays the login access form

The image shows a user login page for the MOMS (Midwifery Outcomes Management System). The page has a light blue background. At the top center, the logo "MOMS" is displayed in blue, with the full name "Midwifery Outcomes Management System" in a smaller font below it. In the center of the page is a white rounded rectangle containing the "Sign In" form. The form has the title "Sign In" and a subtitle "Enter your username and password." Below this are two input fields: "Username" and "Password". At the bottom of the form is a blue button labeled "Sign In". At the very bottom of the white form area, there is a small copyright notice: "© 2026 Midwifery Outcomes Management System".

*Figure 20: User Login Page*

## Screen: 2

Dashboard Page - provides an overview of the system and highlights items that need attention

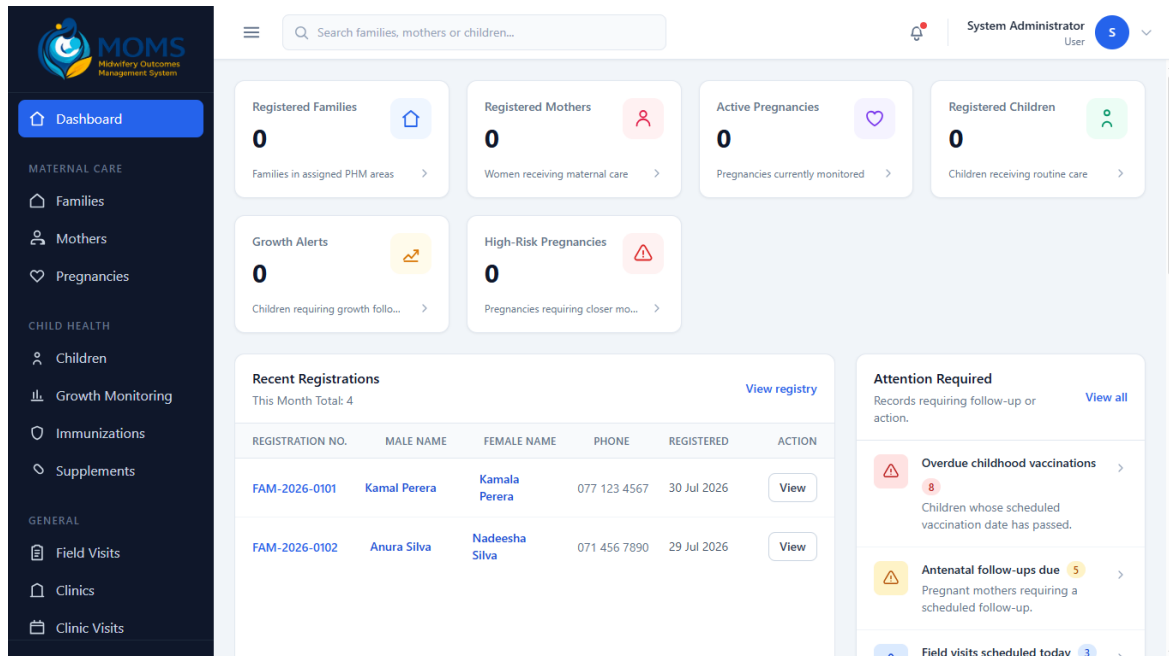


Figure 21: Dashboard Page

## Screen: 3

Create User Page - creates a new user for the system

The screenshot shows the MOMS Create User page. The form is titled 'Create User' and includes a 'Back to Users' button. The form is divided into three sections: Account Information, User Details, and a section for Role and Status. The Account Information section contains fields for Username, Full Name, Password, and Confirm Password. The User Details section contains a dropdown for Role (currently set to 'Select Role') and a dropdown for Status (currently set to 'Active').

| Account Information  |                      |
|----------------------|----------------------|
| Username             | Full Name            |
| <input type="text"/> | <input type="text"/> |
| Password             | Confirm Password     |
| <input type="text"/> | <input type="text"/> |

| User Details |        |
|--------------|--------|
| Role         | Status |
| Select Role  | Active |

Figure 22: Create User Page

## Screen: 4

Family Profile Page - shows details of a single family

The screenshot displays the Family Profile Page for family FAM000014. The left sidebar contains navigation links for Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The main content area shows the family's status as 'Active', registered on 04 Mar 2023, with family ID #0014 and 4 registered members. Below this, four summary cards show: Family members (4), Children under 5 (1), Total pregnancies (2), and Registered Children (0). The 'Family information' section lists the address as No.72 Uluwitike Galle and the PHM area as Galle Town. The 'Family members' section shows a table with one member, Nilmini Silva, a 32-year-old mother, with no active records. An 'Add member' button is present.

Figure 23: Family Profile Page

## Screen: 5

Family Registry Page - shows a list of families

The screenshot displays the Family Registry Page. The left sidebar is identical to the previous screen. The main content area shows the 'Family Registry' section with a search bar and filters. Below the search bar, there is a table listing 5 registered families. The table has columns for Registration No., Address, PHM Area, Registration Date, Status, and Actions. All families are currently 'Active' and located in Galle Town.

| REGISTRATION NO. | ADDRESS                       | PHM AREA   | REGISTRATION DATE | STATUS | ACTIONS |
|------------------|-------------------------------|------------|-------------------|--------|---------|
| FAM000001        | No.12 Hospital Road Galle     | Galle Town | 2023-01-05        | Active |         |
| FAM000002        | No.45 Lighthouse Street Galle | Galle Town | 2023-01-08        | Active |         |
| FAM000003        | No.28 Wakwella Road Galle     | Galle Town | 2023-01-10        | Active |         |
| FAM000004        | No.77 Mahamodara Road Galle   | Galle Town | 2023-01-12        | Active |         |
| FAM000005        | No.16 Karapitiya Road Galle   | Galle Town | 2023-01-18        | Active |         |

Figure 24: Family Registry Page

## Screen: 6

Register New Family Member Page - displays new family member registration form

The screenshot shows the 'Register Family Member' page in the MOMS (Midwifery Outcomes Management System) interface. The left sidebar contains navigation links for Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The main content area has a breadcrumb trail: Dashboard > Families > FAM000020 > Add Member. The page title is 'Register Family Member', and the 'Members' tab is selected. The 'Family Member Registration Form' includes the following fields: Full name (First name, Middle name, Last name), Date of birth (mm/dd/yyyy), Gender (Select Gender), NIC number (901234567V), Contact number (Example: 0771234567), Email address (example@domain.com), Marital status (Select marital status), Occupation, and Education level (Select education level). Each field has a placeholder or example value and a small icon for date or gender selection.

Figure 25: Register New Family Member Page

## Screen: 7

Update Family Member Details Page - updates an already created family member that belong to a family

The screenshot shows the 'Edit Family Member' page in the MOMS system. The left sidebar is identical to the previous screen. The breadcrumb trail is: Dashboard > Families > FAM000001 > Members > 1 > Edit Member. The page title is 'Edit Family Member', and the 'Members' tab is selected. The 'Update Family Member' form contains the following fields: Full name (First name: Nadeesha, Middle name, Last name: Perera), Date of birth (04/12/1990), Gender (Female), NIC number (908451236V), Contact number (775123401), Email address (nadeesha.perera1@example.com), Marital status (Married), Occupation, and Education level. Each field is pre-filled with data for an existing family member.

Figure 26: Update Family Member Details Page

## Screen: 8

User Management Access Grant Page - provides user access to the system

The screenshot shows the 'User Management' page in the MOMS system. The left sidebar contains navigation links for Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The main content area has a search bar at the top with the text 'Search families, mothers or children...'. Below this is a 'Users' section with the title 'User Management' and a subtitle 'Manage application users and their roles.' A '+ Add User' button is in the top right. A search filter section includes a 'Search users...' input, 'All Roles' and 'All Status' dropdowns, and a 'Search' button. Below this is a table with columns: ID, Full Name, Username, Role, Status, and Actions. The table contains one row for the 'System Administrator' user, whose status is 'Inactive'. The 'Actions' column has 'View', 'Edit', and 'Delete' buttons. At the bottom, it shows 'Total Users: 1' and a 'Pagination' link. The footer text is 'Midwifery Outcomes Management System | Version 1.0'.

| ID | Full Name            | Username | Role          | Status   | Actions                                                          |
|----|----------------------|----------|---------------|----------|------------------------------------------------------------------|
| 1  | System Administrator | admin    | Administrator | Inactive | <a href="#">View</a> <a href="#">Edit</a> <a href="#">Delete</a> |

Figure 27: User Management Access Grant Page

## Screen: 9

Families Overview Page - provide a summary detail and useful statistics

The screenshot shows the 'Families Overview' page in the MOMS system. The left sidebar is the same as in Figure 27. The main content area has a search bar at the top with the text 'Search families, mothers or children...'. Below this is a 'Families Overview' section with tabs for 'Overview', 'Registry', 'Register', and 'Reports'. The 'Overview' tab is active. A 'Families Overview' title is followed by 'Open Registry' and 'Register Family' buttons. Below this are four summary cards: 'Registered Families' (126), 'Active Families' (125), 'Inactive Families' (1), and 'Follow-ups Due' (0). Below the cards is a table with columns: REGISTRATION NO., ADDRESS, PHM AREA, and STATUS. The table contains four rows of family data.

| REGISTRATION NO. | ADDRESS                       | PHM AREA     | STATUS |
|------------------|-------------------------------|--------------|--------|
| 1                | No.12 Hospital Road Galle     | Not assigned | Active |
| 2                | No.45 Lighthouse Street Galle | Not assigned | Active |
| 3                | No.28 Wakwella Road Galle     | Not assigned | Active |
| 4                | No.77 Mahamodara Road Galle   | Not assigned | Active |

Figure 28: Families Overview Page

## Screen: 10

### Family Household Page - home details

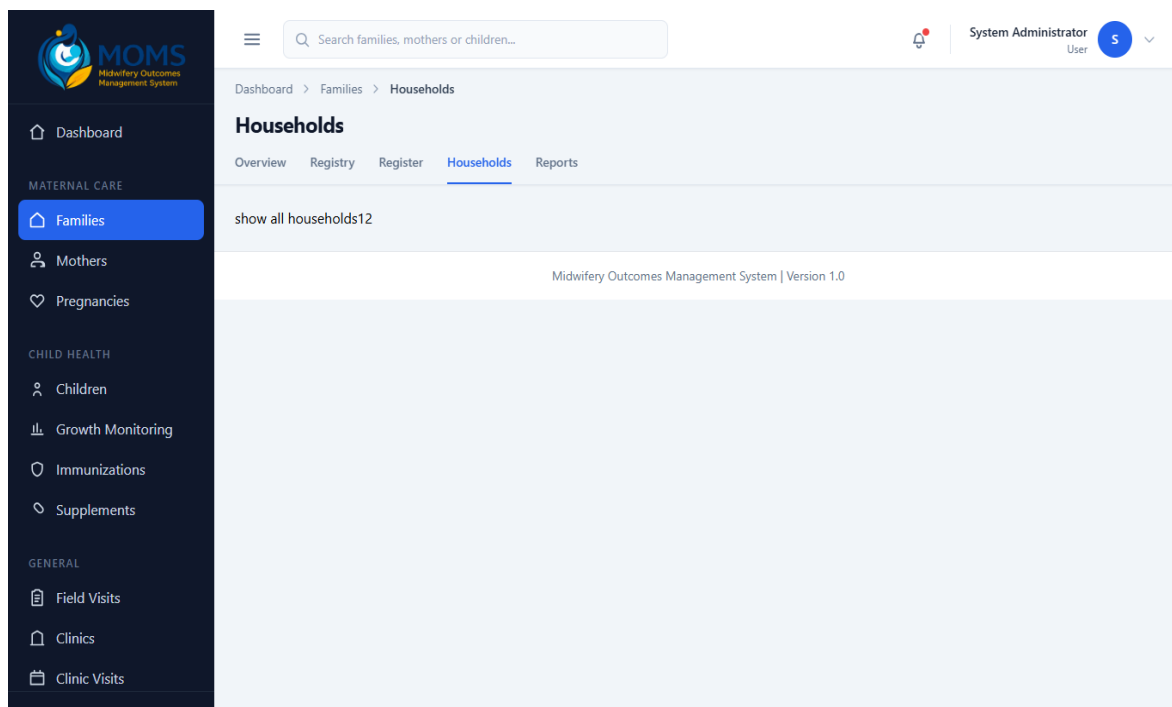


Figure 29: Family Household Page

## Screen: 11

### Members of a Family Page - lists family members belonging to a single family

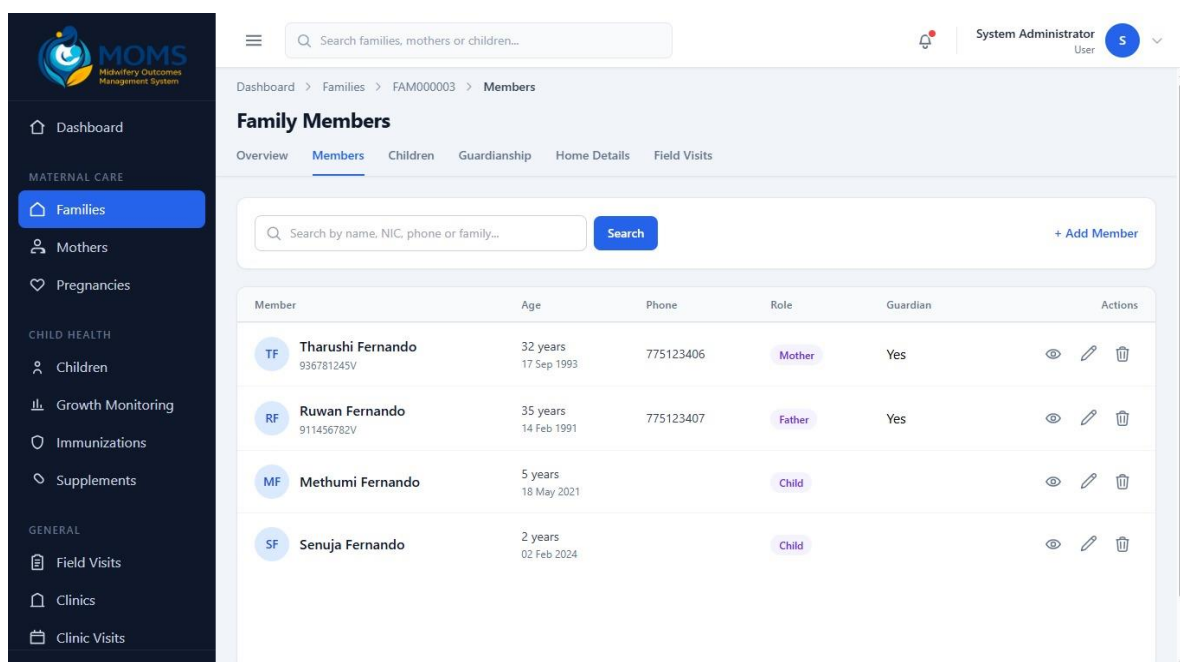


Figure 30: Members of a Family Page

## Screen: 12

Family Member Profile Page - shows details about individual member

The screenshot shows the 'Family Member Profile' page for a user named 'Kasun Perera'. The page is part of the 'MOMS Midwifery Outcomes Management System'. The left sidebar contains navigation links for Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The main content area has a breadcrumb trail: Dashboard > Families > FAM000001 > Members > Kasun Perera. Below the breadcrumb is the title 'Family Member Profile' and tabs for Overview, Members (selected), Children, Guardianship, Home Details, and Field Visits. A 'Back to family' link and an 'Edit member' button are also present. The profile card for 'Kasun Perera' (Father) includes a profile picture and the following details:

| Personal information |            |               |
|----------------------|------------|---------------|
| Full name            | NIC number | Date of birth |
| Kasun                | 881231456V | 23 Jan 1988   |
| Age                  | Gender     | Family role   |
| ---                  | Male       | Father        |

| Contact information |                           |
|---------------------|---------------------------|
| Phone               | Email                     |
| 775123402           | kasun.perera1@example.com |
| ---                 | ---                       |

Figure 31: Family Member Profile Page

## Screen: 13

Mothers' Overview Page - shows summary view of all mothers

The screenshot shows the 'Mothers Overview' page. The left sidebar is the same as in Figure 31, but the 'Mothers' link under MATERNAL CARE is selected. The main content area has a breadcrumb trail: Dashboard > Mothers. Below the breadcrumb is the title 'Mother Overview' and tabs for Overview (selected), Registry, Register Mother, Pregnancies, Supplements, and Reports. There are 'Open Registry' and 'Register Mother' buttons. A summary section displays four key metrics:

- Registered Mothers: 39
- Currently Pregnant: 11
- High Risk: 0
- Deliveries Due: 0 (Within the next 30 days)

| MOTHER                 | NIC        | FAMILY    | PHM AREA   | MATERNAL STATUS | RISK | ACTION                       |
|------------------------|------------|-----------|------------|-----------------|------|------------------------------|
| Chamari Perera         | 927854361V | FAM000038 | Fort       | Pregnant        | Low  | <a href="#">View profile</a> |
| Chathurika Jayawardena | 917654892V | FAM000022 | Galle Town | Registered      | ---  | <a href="#">View profile</a> |
| Dilani Perera          | 938764215V | FAM000044 | Fort       | Registered      | ---  | <a href="#">View profile</a> |
| Dilani Silva           | 927654321V | FAM000002 | Galle Town | Registered      | ---  | <a href="#">View profile</a> |
| Dinushi Jayasinghe     | 899876543V | FAM000016 | Galle Town | Registered      | ---  | <a href="#">View profile</a> |

Figure 32: Mothers' Overview Page

## Screen: 14

Mothers' Registry Page - shows mothers registry

Dashboard > Mothers > Registry

### Mother Registry

Overview **Registry** Register Mother Pregnancies Supplements Reports

Search name, NIC or family number

| MOTHER                               | NIC        | FAMILY    | CONTACT    | PREGNANCIES | CURRENT STATUS | ACTION               |
|--------------------------------------|------------|-----------|------------|-------------|----------------|----------------------|
| Chamari Perera<br>Fort               | 927854361V | FAM000038 | 775123511  | 2           | Pregnant       | <a href="#">View</a> |
| Chathurika Jayawardena<br>Galle Town | 917654892V | FAM000022 | 775123452  | 1           | Registered     | <a href="#">View</a> |
| Dilani Perera<br>Fort                | 938764215V | FAM000044 | 775123526  | 1           | Registered     | <a href="#">View</a> |
| Dilani Silva<br>Galle Town           | 927654321V | FAM000002 | 775123403  | 2           | Registered     | <a href="#">View</a> |
| Dinushi Jayasinghe<br>Galle Town     | 899876543V | FAM000016 | 775123437  | 1           | Registered     | <a href="#">View</a> |
| erg drfgsvwer QWDFe                  | 123123123V | FAM000126 | 0772340561 | 0           | Registered     | <a href="#">View</a> |

Figure 33: Mother Registry Page

## Screen: 15

Pregnancies Overview Page - displays summary details of all pregnancies

Dashboard > Pregnancies

### Pregnancies

Overview **Registry** Register Pregnancy High Risk Expected Deliveries visits and follow-ups Birth Outcomes Reports

Pregnancies

Active Pregnancies  
**12**

High Risk  
**0**

Deliveries Due  
**0**  
Within 30 days

Delivered  
**42**

| MOTHER                 | REGISTRATION DATE | EXPECTED DELIVERY | RISK     | STATUS  |
|------------------------|-------------------|-------------------|----------|---------|
| Harshani Gunasekara    | 2023-06-05        | 2024-01-05        | Low      | Ongoing |
| Nadeesha Silva         | 2023-06-10        | 2024-01-10        | Low      | Ongoing |
| Shashikala Peris       | 2023-07-01        | 2024-01-30        | Low      | Ongoing |
| Shashikala Peris       | 2023-07-11        | 2024-02-09        | Low      | Ongoing |
| Ishani rfr Jayawardena | 2023-07-20        | 2024-02-19        | Moderate | Ongoing |

Figure 34: Pregnancies Overview Page

## Screen: 16

Pregnancy Registry Page - lists all registered pregnancies

The screenshot shows the 'Pregnancies' page in the MOMS (Midwifery Outcomes Management System) interface. The left sidebar contains navigation links for Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The main content area has a search bar and a user profile for 'System Administrator'. Below the search bar, there's a breadcrumb trail 'Dashboard > Pregnancies' and a title 'Pregnancies'. A navigation bar includes 'Overview' (selected), 'Registry', 'Register Pregnancy', 'High Risk', 'Expected Deliveries', 'visits and follow-ups', 'Birth Outcomes', and 'Reports'. A summary section shows four key metrics: Active Pregnancies (12), High Risk (0), Deliveries Due (0, within 30 days), and Delivered (42). Below this is a table listing registered pregnancies with columns for Mother, Registration Date, Expected Delivery, Risk, and Status.

| MOTHER                 | REGISTRATION DATE | EXPECTED DELIVERY | RISK     | STATUS  |
|------------------------|-------------------|-------------------|----------|---------|
| Harshani Gunasekara    | 2023-06-05        | 2024-01-05        | Low      | Ongoing |
| Nadeesha Silva         | 2023-06-10        | 2024-01-10        | Low      | Ongoing |
| Shashikala Peris       | 2023-07-01        | 2024-01-30        | Low      | Ongoing |
| Shashikala Peris       | 2023-07-11        | 2024-02-09        | Low      | Ongoing |
| Ishani rfr Jayawardena | 2023-07-20        | 2024-02-19        | Moderate | Ongoing |

Figure 35: Pregnancy Registry Page

## Screen: 17

Register Pregnancy Page - records new pregnancy details

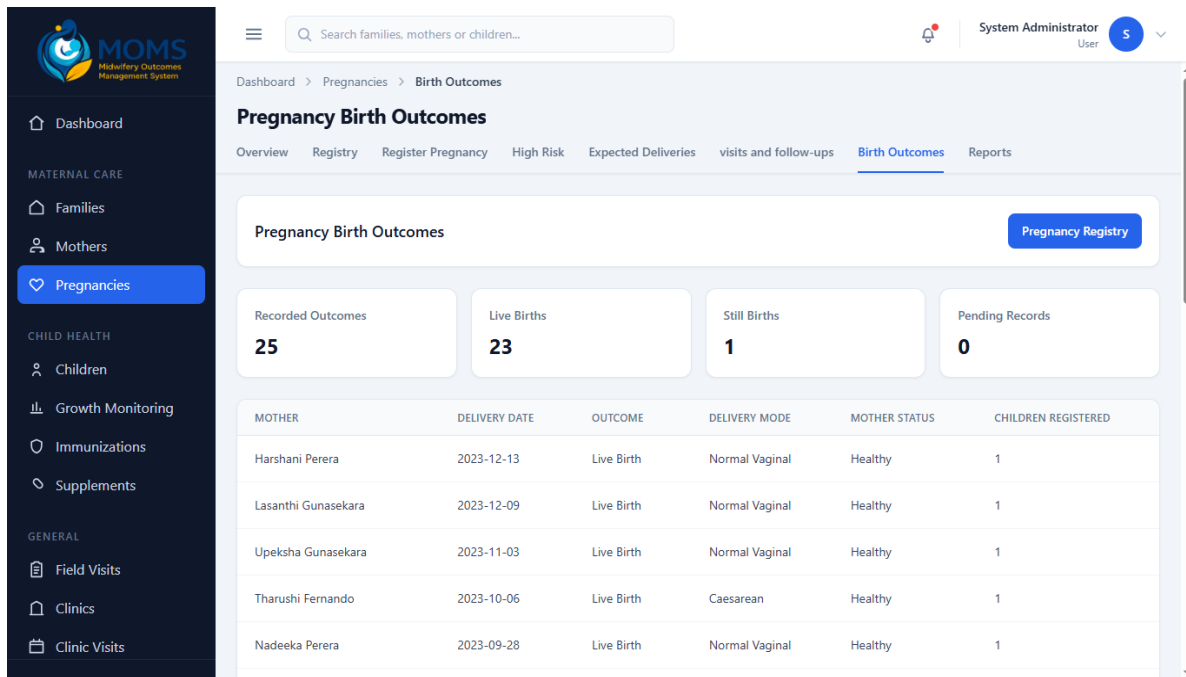
The screenshot shows the 'Register Pregnancy' page in the MOMS system. The left sidebar is identical to the previous screen. The main content area has a breadcrumb trail 'Dashboard > Pregnancies > Register Pregnancy' and a title 'Register Pregnancy'. A navigation bar includes 'Overview', 'Registry', 'Register Pregnancy' (selected), 'High Risk', 'Expected Deliveries', 'visits and follow-ups', 'Birth Outcomes', and 'Reports'. The form is titled 'Pregnancy registration' and contains several input fields for recording pregnancy details.

| Pregnancy registration              |                                      |                                       |
|-------------------------------------|--------------------------------------|---------------------------------------|
| Mother<br>Select registered mother  |                                      |                                       |
| Registered date<br>08/05/2026       | Booking date<br>mm/dd/yyyy           | Pregnancy age at registration (weeks) |
| Last menstrual period<br>mm/dd/yyyy | Expected delivery date<br>mm/dd/yyyy | Blood pressure<br>120/80              |
| Gravida<br>0                        | Para<br>0                            | Abortions<br>0                        |
| Living children<br>0                | BMI at booking                       | Hemoglobin (g/dL)                     |
| Pregnancy type                      | Risk status                          | Current status                        |

Figure 36: Register Pregnancy Page

## Screen: 18

Birth Outcomes Page - shows recorded outcomes for pregnancies



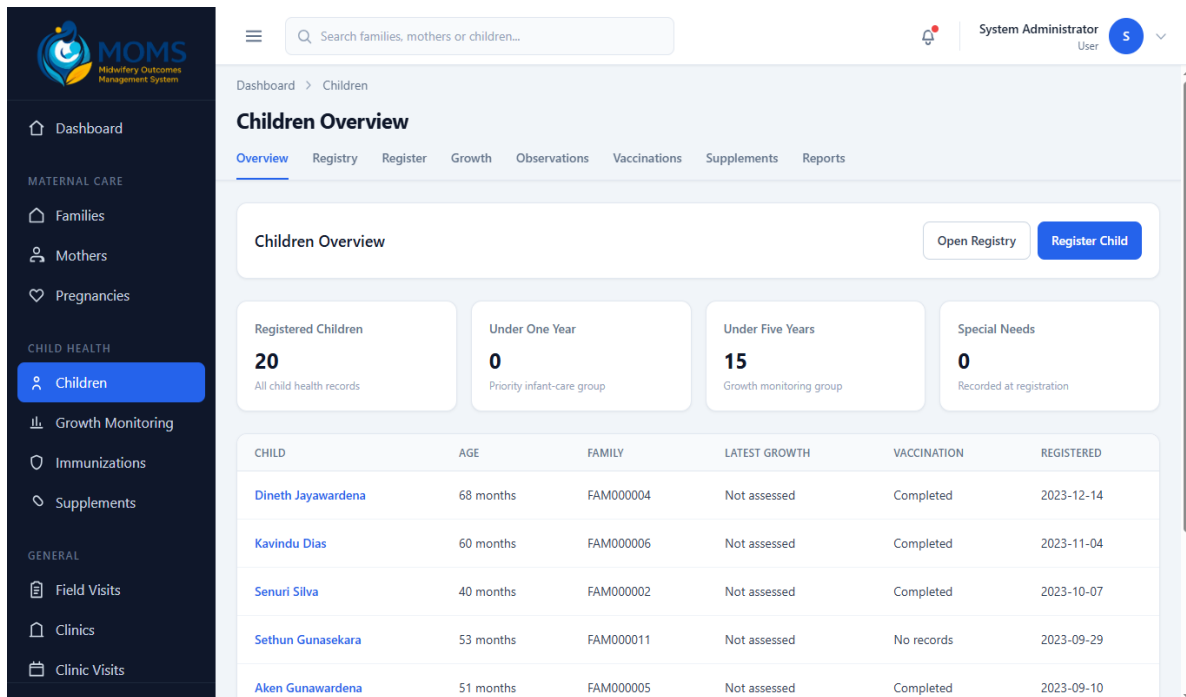
The screenshot displays the 'Pregnancy Birth Outcomes' page in the MOMS system. The left sidebar contains navigation links for Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The main content area shows a search bar, a breadcrumb trail (Dashboard > Pregnancies > Birth Outcomes), and a title 'Pregnancy Birth Outcomes'. Below the title are tabs for Overview, Registry, Register Pregnancy, High Risk, Expected Deliveries, visits and follow-ups, Birth Outcomes (selected), and Reports. A 'Pregnancy Registry' button is in the top right. The summary section includes four cards: Recorded Outcomes (25), Live Births (23), Still Births (1), and Pending Records (0). A table below lists birth records with columns: MOTHER, DELIVERY DATE, OUTCOME, DELIVERY MODE, MOTHER STATUS, and CHILDREN REGISTERED.

| MOTHER              | DELIVERY DATE | OUTCOME    | DELIVERY MODE  | MOTHER STATUS | CHILDREN REGISTERED |
|---------------------|---------------|------------|----------------|---------------|---------------------|
| Harshani Perera     | 2023-12-13    | Live Birth | Normal Vaginal | Healthy       | 1                   |
| Lasanthi Gunasekara | 2023-12-09    | Live Birth | Normal Vaginal | Healthy       | 1                   |
| Upeksha Gunasekara  | 2023-11-03    | Live Birth | Normal Vaginal | Healthy       | 1                   |
| Tharushi Fernando   | 2023-10-06    | Live Birth | Caesarean      | Healthy       | 1                   |
| Nadeeka Perera      | 2023-09-28    | Live Birth | Normal Vaginal | Healthy       | 1                   |

Figure 37: Birth Outcomes Page

## Screen: 19

Children Overview Page - shows Summary details of children below 5 years of age



The screenshot displays the 'Children Overview' page in the MOMS system. The left sidebar contains navigation links for Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The main content area shows a search bar, a breadcrumb trail (Dashboard > Children), and a title 'Children Overview'. Below the title are tabs for Overview (selected), Registry, Register, Growth, Observations, Vaccinations, Supplements, and Reports. 'Open Registry' and 'Register Child' buttons are in the top right. The summary section includes four cards: Registered Children (20), Under One Year (0), Under Five Years (15), and Special Needs (0). A table below lists child records with columns: CHILD, AGE, FAMILY, LATEST GROWTH, VACCINATION, and REGISTERED.

| CHILD              | AGE       | FAMILY    | LATEST GROWTH | VACCINATION | REGISTERED |
|--------------------|-----------|-----------|---------------|-------------|------------|
| Dineth Jayawardena | 68 months | FAM000004 | Not assessed  | Completed   | 2023-12-14 |
| Kavindu Dias       | 60 months | FAM000006 | Not assessed  | Completed   | 2023-11-04 |
| Senuri Silva       | 40 months | FAM000002 | Not assessed  | Completed   | 2023-10-07 |
| Sethun Gunasekara  | 53 months | FAM000011 | Not assessed  | No records  | 2023-09-29 |
| Aken Gunawardena   | 51 months | FAM000005 | Not assessed  | Completed   | 2023-09-10 |

Figure 38: Children Overview Page

## Screen: 20

Children Registry Page - shows filtered list of children below 5 years of age

The screenshot displays the 'Children Registry' page. On the left is a dark sidebar with navigation options: Dashboard, MATERNAL CARE (Families, Mothers, Pregnancies), CHILD HEALTH (Children, Growth Monitoring, Immunizations, Supplements), and GENERAL (Field Visits, Clinics, Clinic Visits). The 'Children' option is highlighted. The main content area has a top navigation bar with 'Overview', 'Registry', 'Register', 'Growth', 'Observations', 'Vaccinations', 'Supplements', and 'Reports'. Below this is a search bar with the placeholder 'Search name, NIC or family number' and a 'Search' button. A 'Register Child' button is also present. The main table lists children with columns: CHILD, DOB / AGE, FAMILY, MOTHER, GROWTH, and ACTIONS. The data rows are as follows:

| CHILD                                 | DOB / AGE               | FAMILY    | MOTHER                 | GROWTH       | ACTIONS                      |
|---------------------------------------|-------------------------|-----------|------------------------|--------------|------------------------------|
| Sethmi Gunawardena<br>No NIC recorded | 2025-01-18<br>18 months | FAM000005 | Nirosha Jayasinghe     | Not assessed | <a href="#">View profile</a> |
| Sehansa Perera<br>No NIC recorded     | 2024-04-17<br>27 months | FAM000007 | Nilmini Silva          | Not assessed | <a href="#">View profile</a> |
| Senuja Fernando<br>No NIC recorded    | 2024-02-02<br>30 months | FAM000003 | Nilmini Dias           | Not assessed | <a href="#">View profile</a> |
| Sithuli Perera<br>No NIC recorded     | 2024-01-20<br>30 months | FAM000001 | Dilani Silva           | Not assessed | <a href="#">View profile</a> |
| Hasini Jayawardena<br>No NIC recorded | 2023-12-01<br>32 months | FAM000004 | Pabasari Silva         | Not assessed | <a href="#">View profile</a> |
| Dinuka Dias                           | 2023-11-20              | FAM000012 | Chathurika Jayawardena | Not assessed | <a href="#">View profile</a> |

Figure 39: Children Registry Page

## Screen: 21

Child Profile Page - shows general details of the selected child

The screenshot displays the 'Child Profile Page' for 'Dineth Jayawardena'. The sidebar is identical to the previous screen, with 'Children' highlighted. The main content area has a top navigation bar with 'Overview', 'Growth', 'Observations', 'Vaccinations', and 'Supplements'. Below this is a search bar and a 'PHM: Sujeewa A. Perera' user profile. The main profile section shows a blue header with the child's name 'Dineth Jayawardena' and family ID 'FAM000004 - 8 months old'. Below this are four cards: 'Birth Weight' (3.25 kg), 'Growth Status' (Normal), 'Next Vaccine' (OPV), and 'Observations' (5 recorded). The bottom section is divided into two columns: 'Personal and family details' and 'Birth and neonatal details'.

| Personal and family details |                           | Birth and neonatal details      |                                                |
|-----------------------------|---------------------------|---------------------------------|------------------------------------------------|
| DATE OF BIRTH<br>2025-12-13 | GENDER<br>Male            | DELIVERY DATE<br>2025-12-13     | DELIVERY PLACE<br>Teaching Hospital Karapitiya |
| NIC<br>Not recorded         | MOTHER<br>Harshani Perera | DELIVERY MODE<br>Normal Vaginal | GESTATION<br>39 weeks                          |

Figure 40: Child Profile Page

## Screen: 22

Child Growth History Page - shows growth history on dynamic charts for an individual child

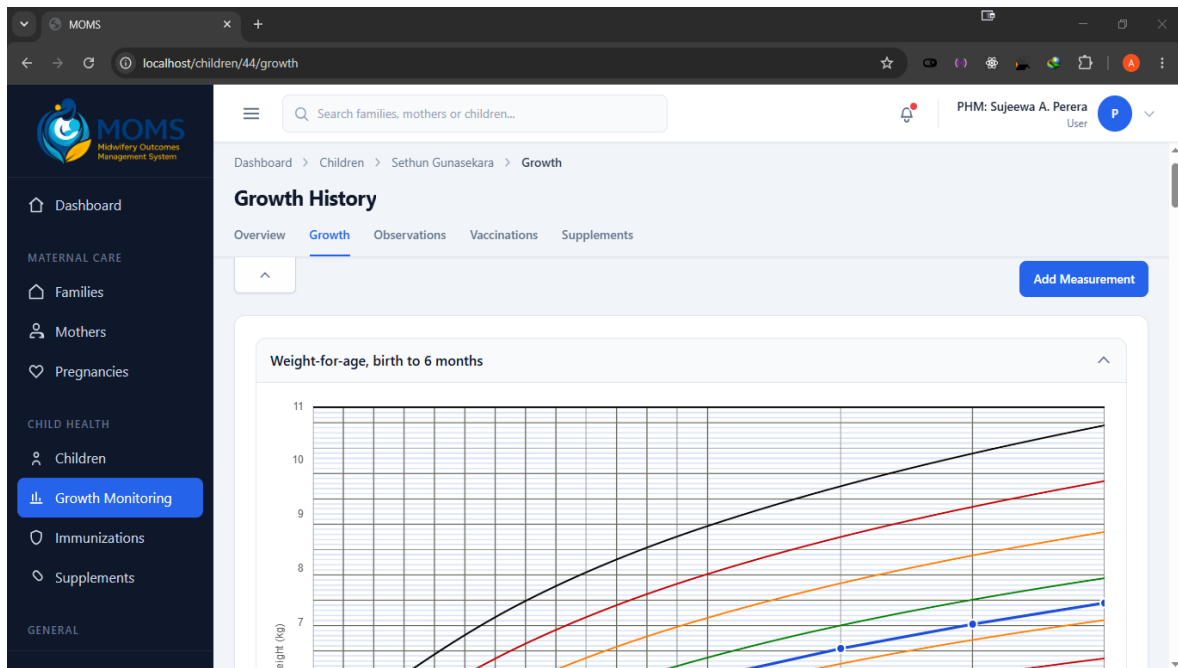


Figure 41.1: Child Growth History Page - part 1



Figure 41.2: Child Growth History Page - part 2

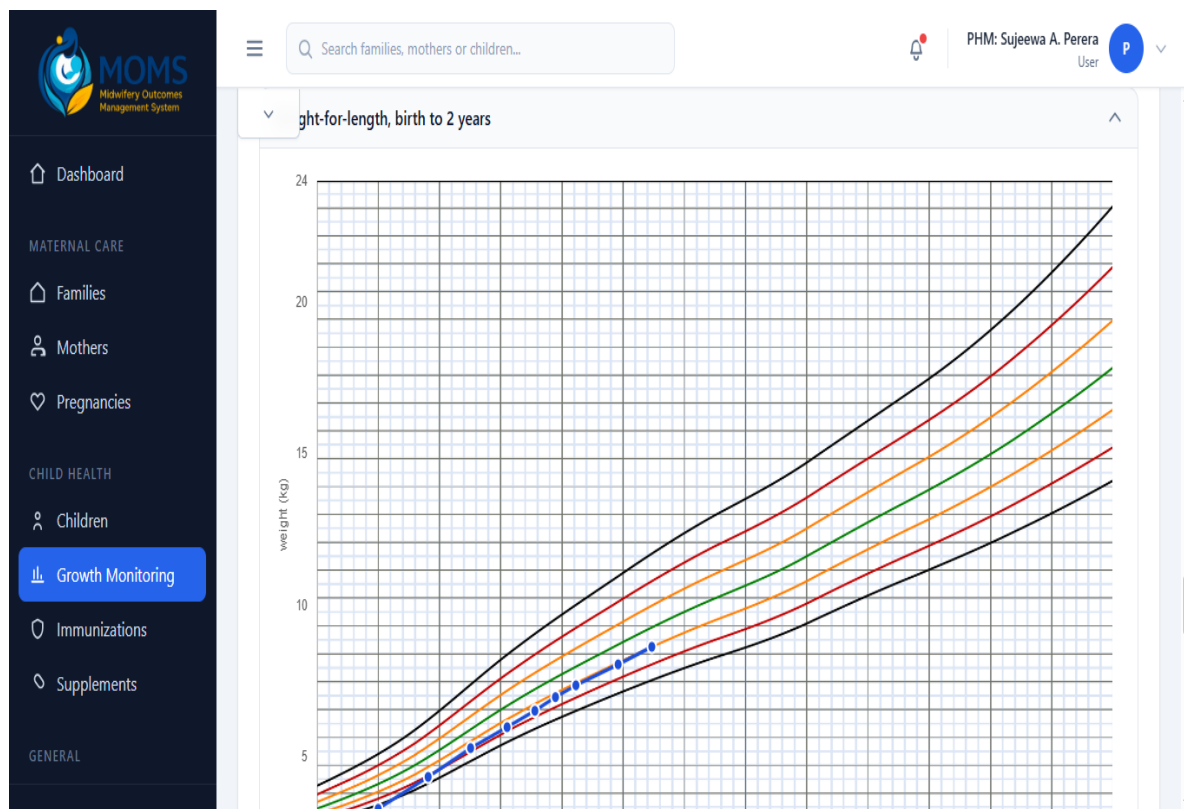


Figure 41.3: Child Growth History page - part 3

### 4.3 Database Design

The database design of the Midwifery Management System was created to organize and manage all required healthcare information properly. The database was implemented using MySQL, a relational database management system to store, update, and retrieve data efficiently. The database contains several important tables to manage different parts of the system.

The proposed midwifery outcomes management system handles heavily structured data that can be clearly normalized into separate entities that have well defined relationships among them. Hence it was essential to join entities with foreign key constraints and thus the relational database management was the database approach that was identified as the most suitable approach for this project.

### 4.3.1 ER Diagram - Overview

An ER diagram is the visual representation of the structure of a relational database which maps the entities and how they relate to each other. The figure 42 shows the entire ER diagram overview and from figure 43.1 to figure 43.7 shows, parts of the ER diagram for each functional section of the system. This was done in order to improve clarity and readability.

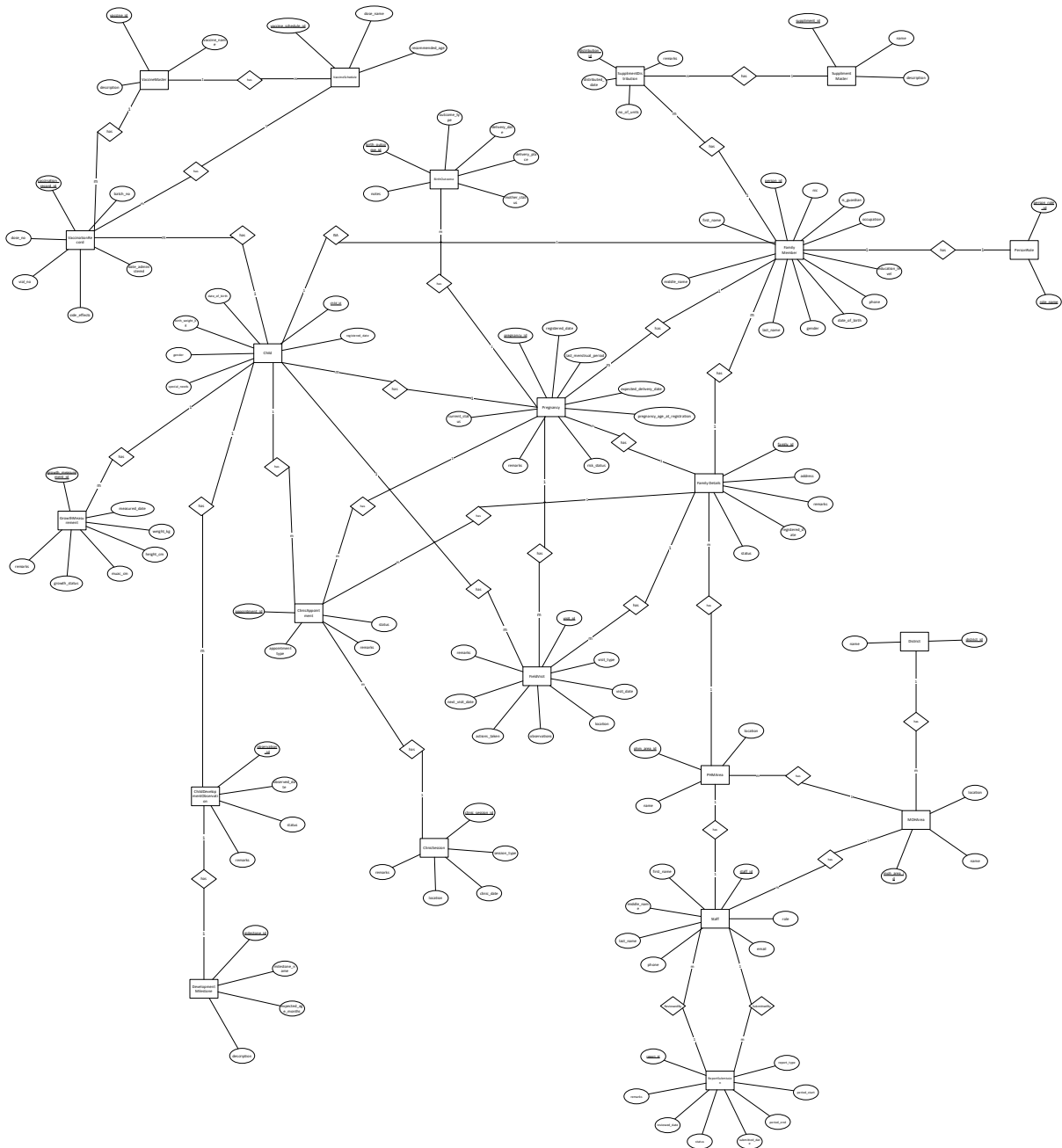


Figure 42: Entity Relationship Diagram Overview

### 4.3.2 ER Diagram - Area and Staff Administration

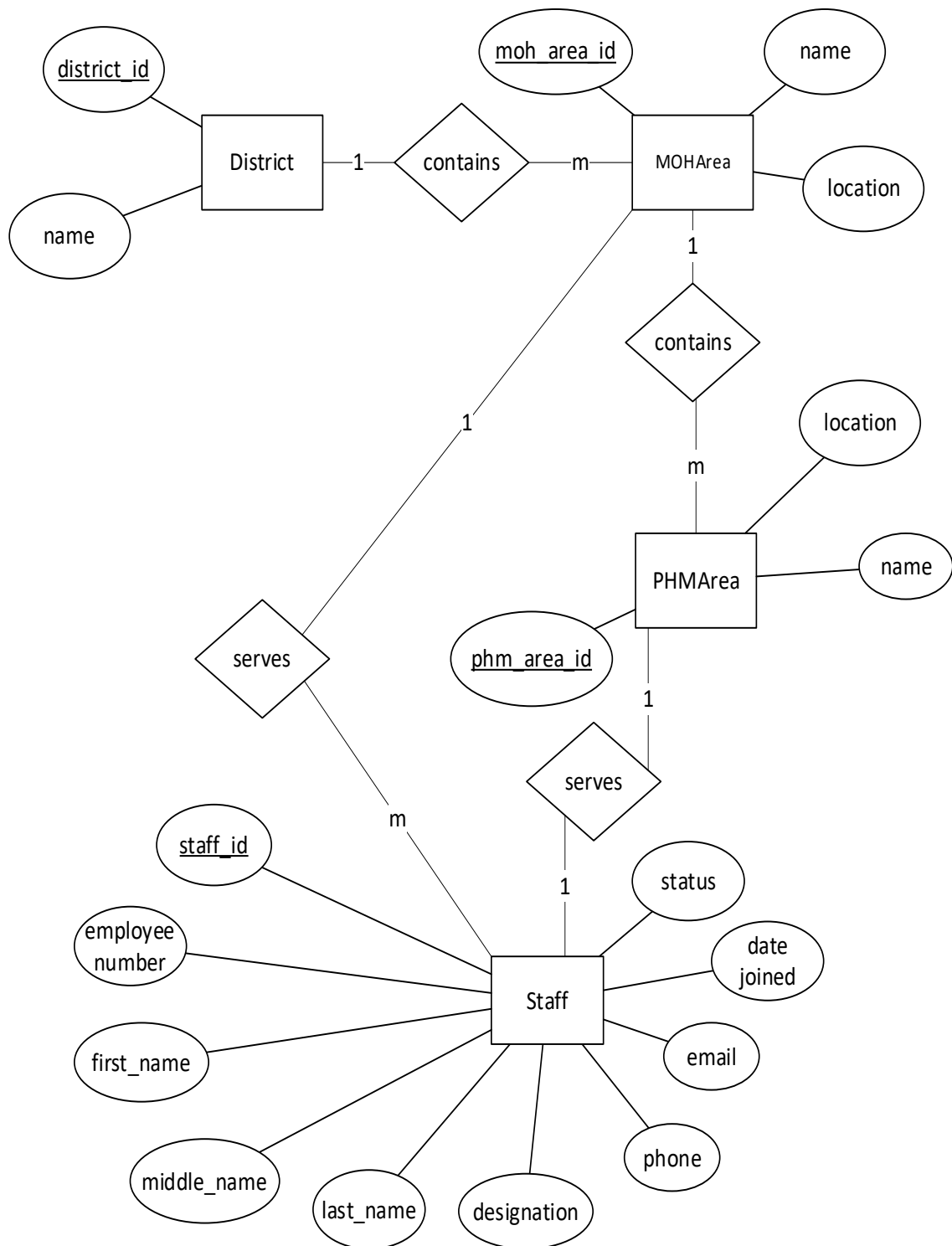


Figure 43.1: ER Diagram - Area and Staff Administration

### 4.3.3 ER Diagram - User Access, Family and Person Registry

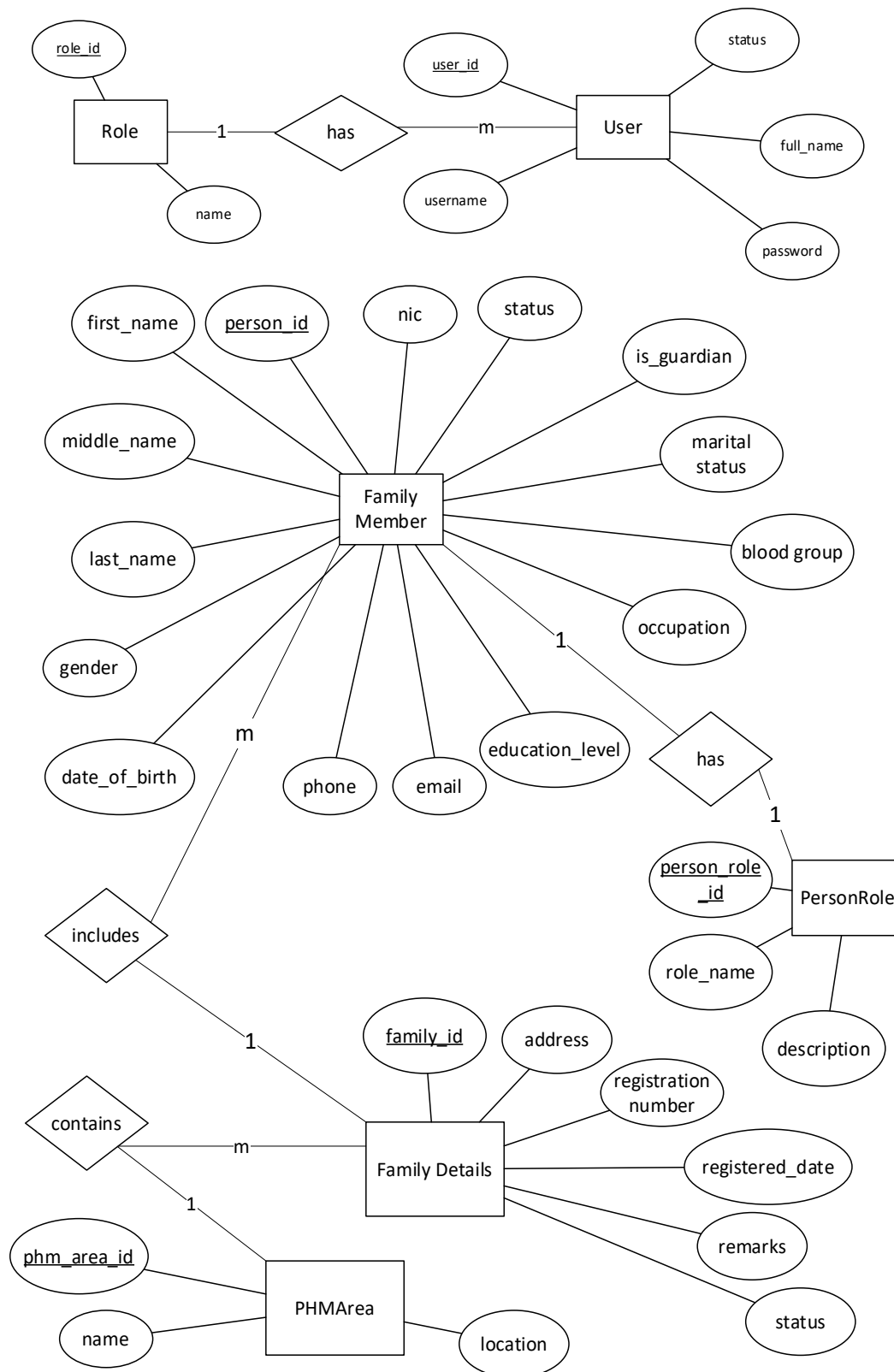


Figure 43.2: ER Diagram - User Access and Person Registry

#### 4.3.4 ER Diagram - Pregnancy Birth and Child Registration

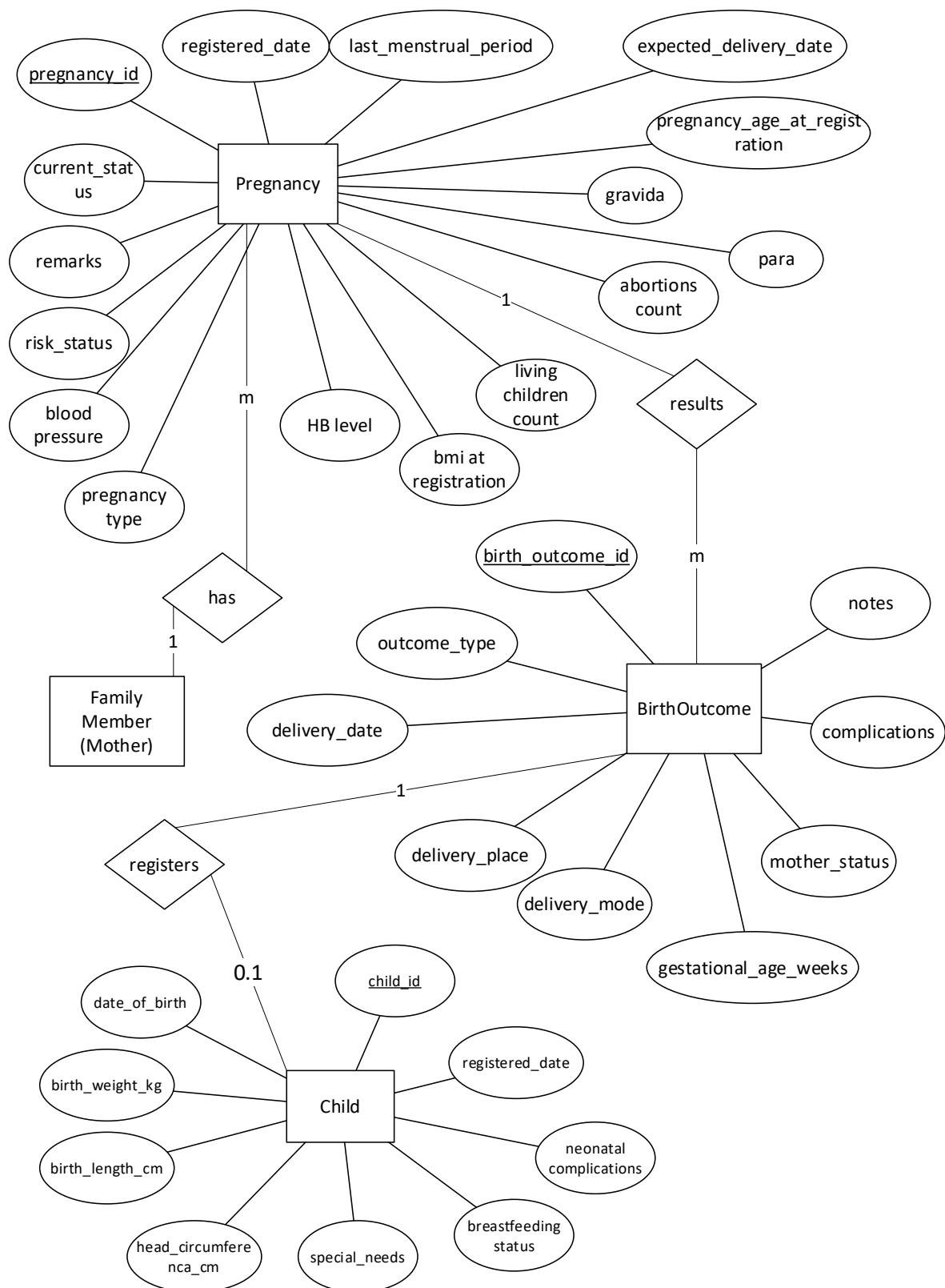


Figure 43.3: ER Diagram - Pregnancy Birth and Child Registration

### 4.3.5 ER Diagram - Child Growth and Development

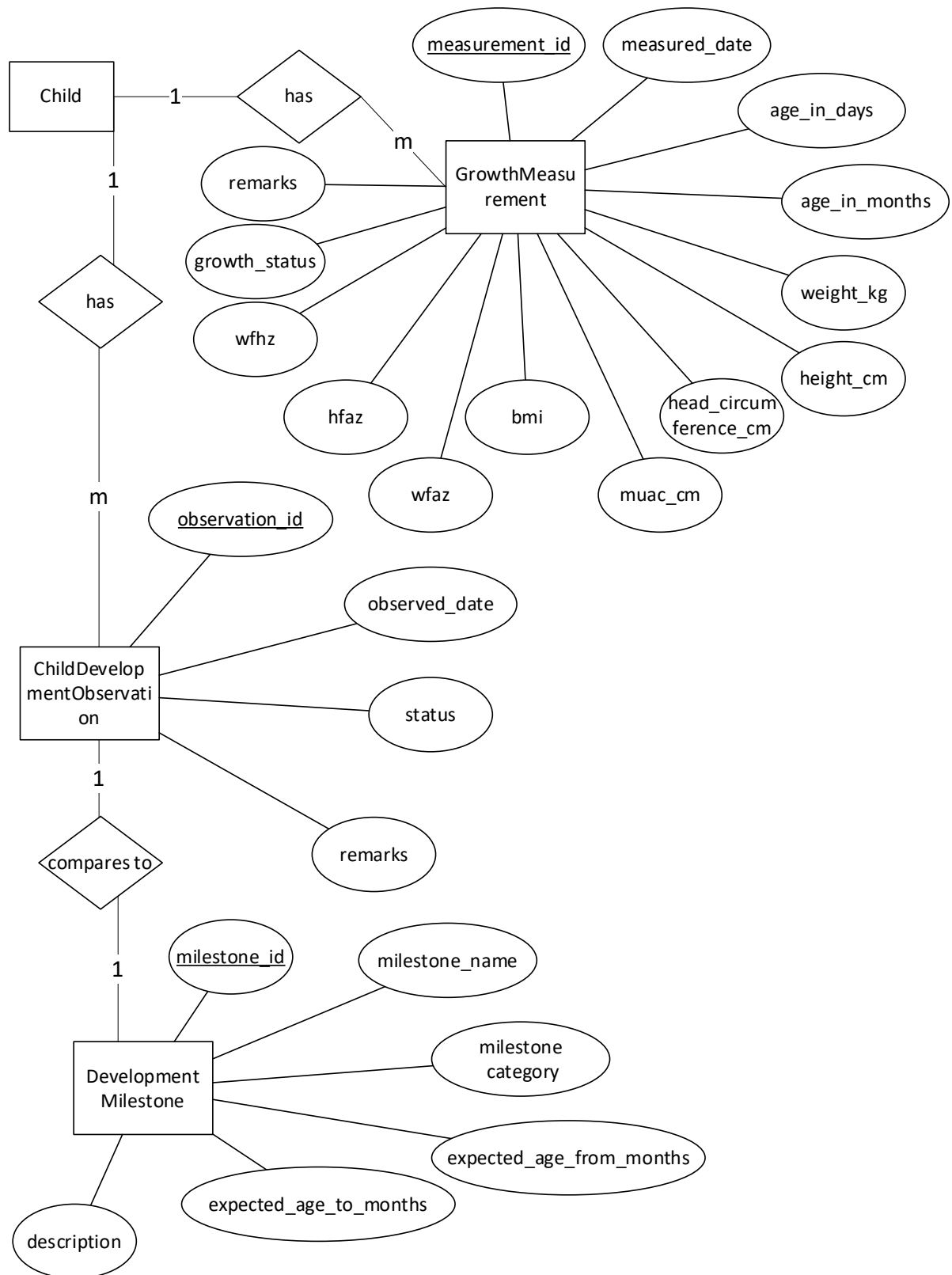


Figure 43.4: ER Diagram - Child Growth and Development

### 4.3.6 ER Diagram - Immunization and Supplementation

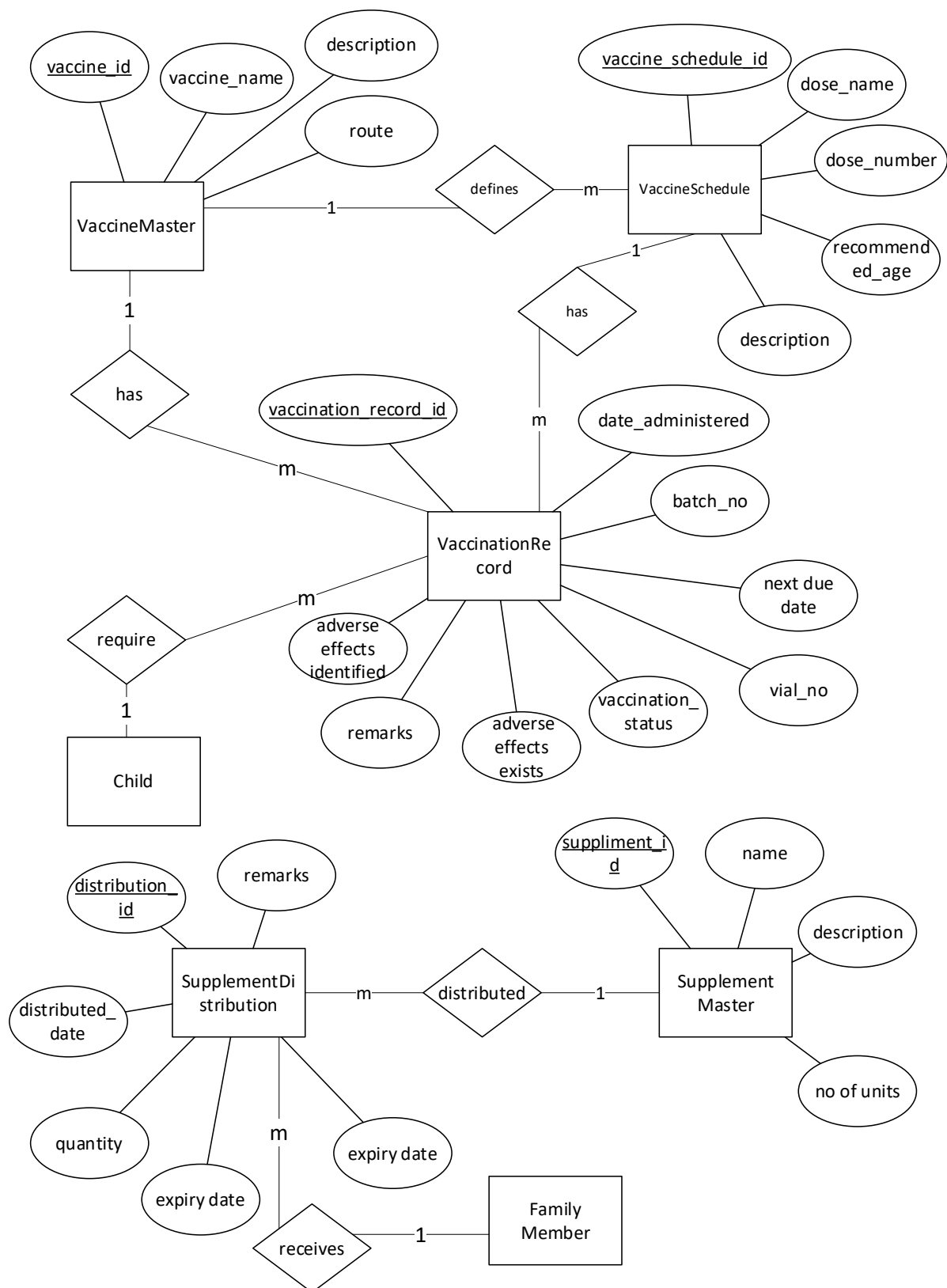


Figure 43.5: ER Diagram - Immunization and Supplementation

#### 4.3.7 ER Diagram - Clinic and Field Visit Activities

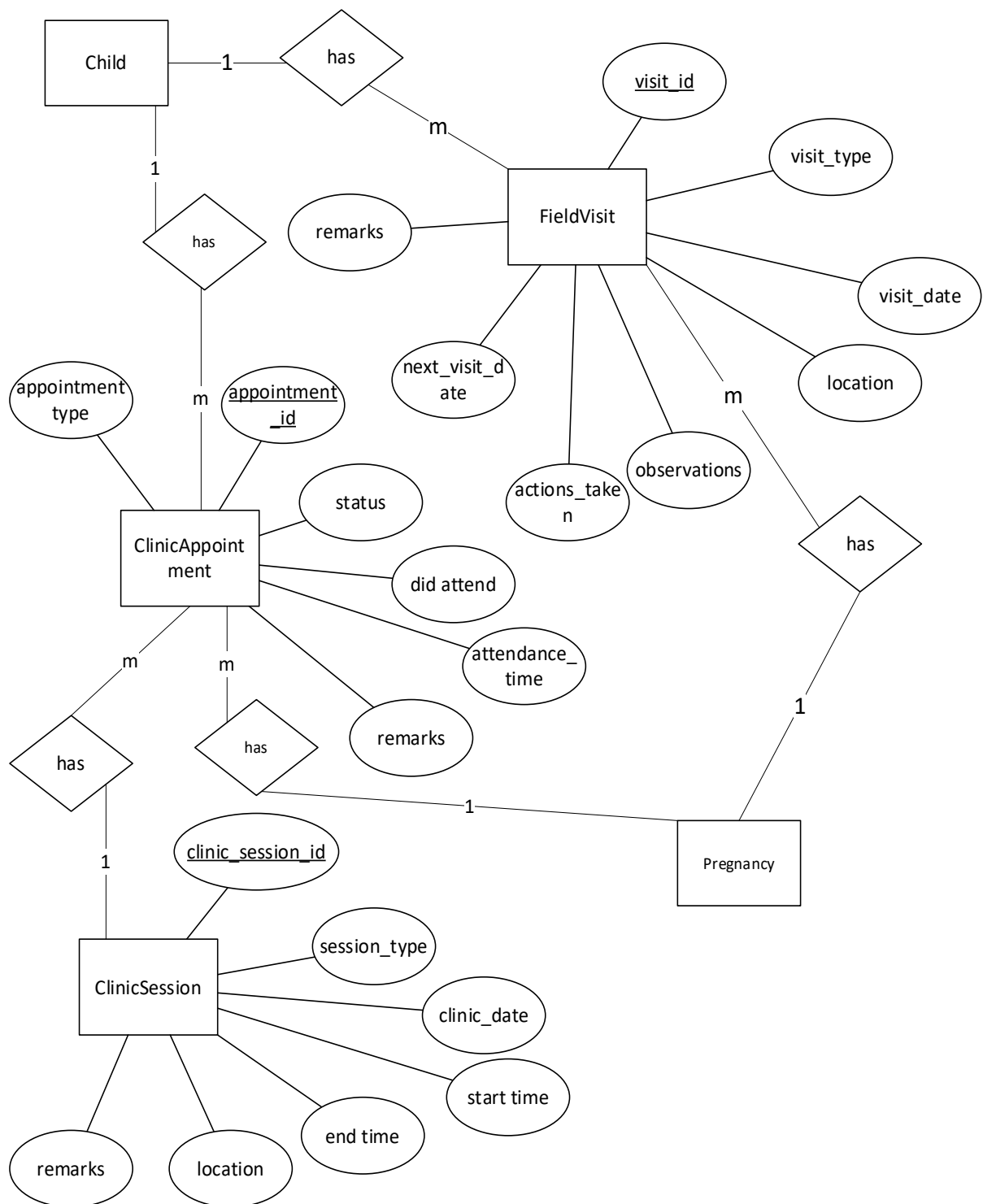


Figure 43.6: ER Diagram - Clinic and Field Visit Activities

### 4.3.8 ER Diagram - Reporting

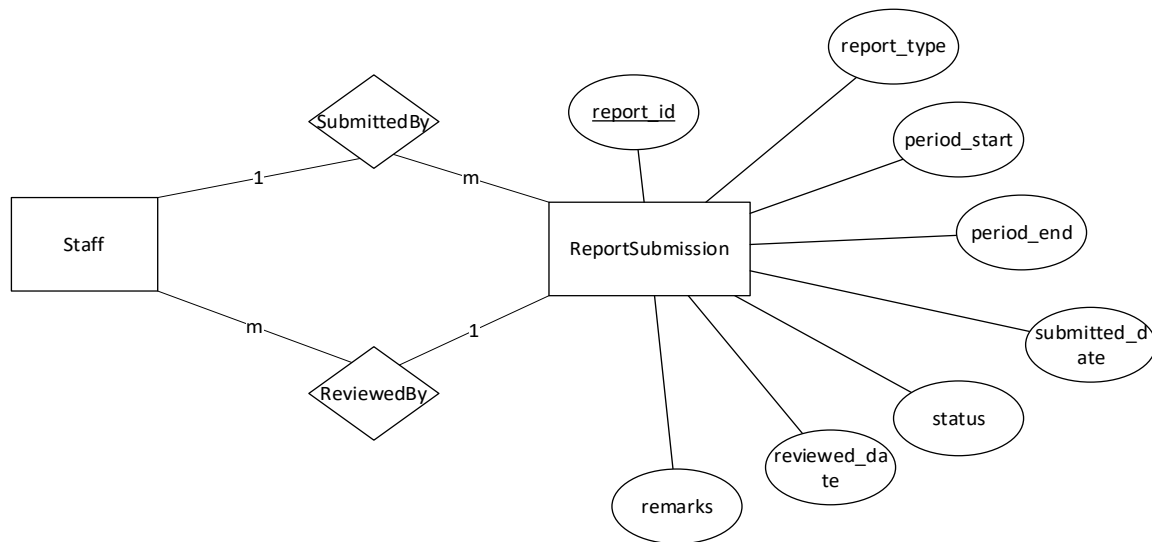


Figure 43.7: ER Diagram – Reporting

## 4.4 Database Tables

### T01: User

Table Name: User

Description: Used to store application user account details, and user credentials

Primary Key: id

References: roles(role\_id)

| Field        | Data Type     | Description                                             |
|--------------|---------------|---------------------------------------------------------|
| Id (PK)      | Integer       | Unique key of the application user account              |
| Username     | Text          | username used to sign in                                |
| Password     | Text          | Secure password                                         |
| full_name    | Text          | Display name of the account holder                      |
| role_id (FK) | Integer       | Application role that determines permissions            |
| Status       | Text          | Whether the account is ACTIVE or INACTIVE               |
| created_at   | Date and time | Date and time when the record was created               |
| updated_at   | Date and time | Date and time when the record was most recently updated |

Table 02: User

## T02: Role

Table Name: Role

Description: Used to store the application access roles that have different permissions

Primary Key: id

References: None

| Field   | Data Type | Description        |
|---------|-----------|--------------------|
| Id (PK) | Integer   | Unique key of role |
| Name    | Text      | Name of role       |

Table 03: Role

## T03: District

Table Name: District

Description: Used to store the administrative districts

Primary Key: district\_id

References: None

| Field            | Data Type     | Description                                             |
|------------------|---------------|---------------------------------------------------------|
| district_id (PK) | Integer       | Unique identifier of the district                       |
| district_name    | Text          | Name of the district                                    |
| created_at       | Date and time | Date and time when the record was created               |
| updated_at       | Date and time | Date and time when the record was most recently updated |

Table 04: District

## T04: MOH Area

Table Name: MOH Area

Description: Stores Medical Officer of Health areas

Primary Key: moh\_area\_id

References: district(district\_id)

| Field            | Data Type | Description                            |
|------------------|-----------|----------------------------------------|
| moh_area_id(PK)  | Integer   | Unique identifier of the MOH area      |
| district_id (FK) | Integer   | District to which the MOH area belongs |
| moh_name         | Text      | Name of the MOH area                   |
| Location         | Text      | Office address                         |

| Field      | Data Type     | Description                                             |
|------------|---------------|---------------------------------------------------------|
| created_at | Date and time | Date and time when the record was created               |
| updated_at | Date and time | Date and time when the record was most recently updated |

Table 05: MOH Area

## T05: PHM Area

Table Name: PHM Area

Description: Stores Public Health Midwife service areas

Primary Key: phm\_area\_id

References: moh\_area(moh\_area\_id)

| Field            | Data Type     | Description                                             |
|------------------|---------------|---------------------------------------------------------|
| phm_area_id(PK)  | Integer       | Unique identifier of the PHM area                       |
| moh_area_id (FK) | Integer       | MOH area to which the PHM area belongs                  |
| phm_area_name    | Text          | Name of the PHM service area                            |
| Location         | Text          | Office location                                         |
| created_at       | Date and time | Date and time when the record was created               |
| updated_at       | Date and time | Date and time when the record was most recently updated |

Table 06: PHM Area

## T06: Staff

Table Name: Staff

Description: Used to store staff details

Primary Key: staff\_id

References: moh\_area(moh\_area\_id), phm\_area(phm\_area\_id)

| Field            | Data Type | Description                           |
|------------------|-----------|---------------------------------------|
| staff_id (PK)    | Integer   | Unique id of the staff member.        |
| employee_number  | Text      | employee number assigned              |
| moh_area_id (FK) | Integer   | MOH area of the staff member          |
| phm_area_id (FK) | Integer   | PHM area assigned to the staff member |
| first_name       | Text      | Staff member's first name             |
| middle_name      | Text      | Staff member's middle name            |
| last_name        | Text      | Staff member's surname                |
| Designation      | Text      | Professional designation              |

| Field       | Data Type     | Description                                    |
|-------------|---------------|------------------------------------------------|
| Phone       | Text          | Contact telephone number                       |
| Email       | Text          | Email address.                                 |
| date_joined | Date          | Date the staff member joined                   |
| Status      | Text          | Current employment status                      |
| created_at  | Date and time | Date and time when the record was created      |
| updated_at  | Date and time | Date and time when the record was last updated |

Table 07: Staff

### T07: Family

Table Name: Family

Description: Used to store registered family details within a PHM area

Primary Key: family\_id

References: phm\_area(phm\_area\_id)

| Field               | Data Type     | Description                                             |
|---------------------|---------------|---------------------------------------------------------|
| family_id (PK)      | Integer       | id of the registered family.                            |
| Phm_area_id (FK)    | Integer       | PHM area of the family                                  |
| Address             | Text          | Residential address of the family                       |
| Registration_number | Text          | Registration number assigned to the family              |
| Registered_date     | Date and time | Family registration date                                |
| Status              | Text          | Whether the family is active, inactive or transferred   |
| Remarks             | Text          | Notes about family and concerns                         |
| Created_at          | Date and time | Date and time when the record was created               |
| Updated_at          | Date and time | Date and time when the record was most recently updated |

Table 08: Family

### T08: Person Role

Table Name: Person Role

Description: Used to group people according to their role in the family

Primary Key: person\_role\_id

References: None

| Field               | Data Type | Description                  |
|---------------------|-----------|------------------------------|
| person_role_id (PK) | Integer   | id of the role in the family |
| role_name           | Text      | Name of the person role      |
| Description         | Text      | Description of the role      |

Table 09: Person Role

## T09: Person

Table Name: Person

Description: User to store the personal details of family members, mothers, children, guardians

Primary Key: person\_id

References: family(family\_id), person\_role(person\_role\_id)

| Field               | Data Type | Description                                         |
|---------------------|-----------|-----------------------------------------------------|
| person_id (PK)      | Integer   | Id of the person in the registry                    |
| family_id (FK)      | Integer   | Registered family to which the person belongs       |
| person_role_id (FK) | Integer   | Id of the role the person belongs to                |
| Nic                 | Text      | National Identity Card number                       |
| first_name          | Text      | Person's first name.                                |
| middle_name         | Text      | Person's middle name                                |
| last_name           | Text      | Person's last name                                  |
| Gender              | Text      | Name of the gender                                  |
| date_of_birth       | Date      | Person's date of birth                              |
| Phone               | Text      | Contact telephone number                            |
| Email               | Text      | Contact email address                               |
| education_level     | Text      | Highest education level                             |
| occupation          | Text      | Current occupation                                  |
| blood_group         | Text      | Recorded blood group                                |
| marital_status      | Text      | Current marital status                              |
| is_guardian         | Boolean   | Whether the person is a guardian of a minor         |
| Status              | Text      | Whether the person is active in the registry or not |

Table 10: Person

## T10: Pregnancy

Table Name: Pregnancy

Description: Used to store pregnancy registration, related data and current status.

Primary Key: pregnancy\_id

References: person(mother\_id)

| Field                 | Data Type | Description                            |
|-----------------------|-----------|----------------------------------------|
| pregnancy_id (PK)     | Integer   | Unique id of the pregnancy             |
| mother_id (FK)        | Integer   | Person id of the pregnant mother       |
| registered_date       | Date      | Date the pregnancy was registered      |
| last_menstrual_period | Date      | First day of the last menstrual period |

| Field                         | Data Type     | Description                                                 |
|-------------------------------|---------------|-------------------------------------------------------------|
| expected_delivery_date        | Date          | Estimated date of delivery                                  |
| pregnancy_age_at_registration | Integer       | Estimated age of the pregnancy                              |
| Gravida                       | Integer       | Total number of pregnancies including the current pregnancy |
| Para                          | Integer       | Number of pregnancies reaching the birth                    |
| Abortions                     | Integer       | Number of previous abortions or miscarriages                |
| living_children               | Integer       | Number of living children                                   |
| bmi_at_booking                | Decimal       | Body mass index measured                                    |
| blood_pressure                | Text          | Blood pressure value recorded                               |
| pregnancy_type                | Text          | One or multiple fetus                                       |
| risk_status                   | Text          | Current pregnancy risk level                                |
| current_status                | Text          | Status of the pregnancy                                     |
| Remarks                       | Text          | Additional observations or risk notes                       |
| created_at                    | Date and time | Date and time when the record was created                   |
| updated_at                    | Date and time | Date and time when the record was last updated              |

Table 11: Pregnancy

## T11: Birth Outcome

Table Name: Birth Outcome

Description: Used to store the conclusion and delivery details of a pregnancy

Primary Key: birth\_outcome\_id

References: pregnancy(pregnancy\_id)

| Field                 | Data Type     | Description                                                    |
|-----------------------|---------------|----------------------------------------------------------------|
| birth_outcome_id (PK) | Integer       | id of the pregnancy outcome record                             |
| pregnancy_id (FK)     | Integer       | Pregnancy the outcome belongs                                  |
| outcome_type          | Text          | Result of the pregnancy (live birth, still birth, miscarriage) |
| delivery_date         | Date          | Date on which delivery or end of pregnancy happened            |
| delivery_place        | Text          | The place where the delivery happened                          |
| delivery_mode         | Text          | Method, child was delivered                                    |
| gestational_age_weeks | Integer       | Completed weeks of gestation when delivery happened            |
| mother_status         | Text          | Mother's condition after the pregnancy outcome                 |
| complications         | Text          | Complications recorded for the outcome                         |
| Notes                 | Text          | Other information about the outcome                            |
| created_at            | Date and time | Date and time when the record was created                      |
| updated_at            | Date and time | Date and time when the record was last updated                 |

Table 12: Birth Outcome

## T12: Child

Table Name: Child

Description: Used to store child-specific birth details linked to a child person and a birth outcome

Primary Key: person\_id

References: person(person\_id), birth\_outcome(birth\_outcome\_id)

| Field                  | Data Type     | Description                                                    |
|------------------------|---------------|----------------------------------------------------------------|
| person_id (FK)         | Integer       | Person record id of child and also the child table primary key |
| birth_outcome_id (PK)  | Integer       | Birth outcome the child belongs to                             |
| birth_weight_kg        | Decimal       | Weight of the child at birth in kilograms                      |
| birth_length_cm        | Decimal       | Length of the child at birth in centimeters                    |
| head_circumference_cm  | Decimal       | Head circumference at birth in centimeters                     |
| special_needs          | Boolean       | Indicates whether special needs have been identified           |
| breastfeeding_status   | Text          | Current feeding method                                         |
| neonatal_complications | Text          | Complications identified during the neonatal period            |
| registered_date        | Date          | Date on which the child was registered                         |
| created_at             | Date and time | Date and time when the record was created.                     |
| updated_at             | Date and time | Date and time when the record was last updated.                |

Table 1: Child

## T13: Development Milestone

Table Name: Development Milestone

Description: Used to store the standard list of child developmental milestones and expected age ranges

Primary Key: milestone\_id

References: None

| Field                   | Data          | Description                                             |
|-------------------------|---------------|---------------------------------------------------------|
| milestone_id (PK)       | Integer       | Unique id of the milestone definition                   |
| milestone_name          | Text          | Name of the ability                                     |
| milestone_category      | Text          | Development area represented                            |
| expected_age_from_month | Integer       | Earliest expected age in months                         |
| expected_age_to_month   | Integer       | Latest expected age in months                           |
| Description             | Text          | Detailed explanation                                    |
| created_at              | Date and time | Date and time when the record was created               |
| updated_at              | Date and time | Date and time when the record was most recently updated |

Table 2: Development Milestone

## T14: Child Development Observation

Table Name: Child Development Observation

Description: Used to record development observations of children

Primary Key: observation\_id

References: child(child\_id), development\_milestone(milestone\_id), staff(observed\_by)

| Field               | Data Type     | Description                                                 |
|---------------------|---------------|-------------------------------------------------------------|
| observation_id (PK) | Integer       | Unique id of the developmental observation                  |
| child_id (FK)       | Integer       | Child id of the observation                                 |
| milestone_id (FK)   | Integer       | Developmental milestone identified during the observation   |
| observed_by         | Integer       | Staff member who recorded the assessment                    |
| observation_date    | Date          | Date on which the milestone was observed                    |
| observation_status  | Text          | Whether the milestone was achieved, delayed or not achieved |
| Remarks             | Text          | Notes or supporting observations                            |
| created_at          | Date and time | Date and time when the record was created                   |
| updated_at          | Date and time | Date and time when the record was most recently updated     |

Table 3: Child Development Observation

## T15: Growth Measurement

Table Name: Growth Measurement

Description: Store child growth measurements, calculated z-scores, and growth state

Primary Key: measurement\_id

References: child(child\_id), staff(measured\_by)

| Field                     | Data Type | Description                                |
|---------------------------|-----------|--------------------------------------------|
| measurement_id (PK)       | Integer   | Unique id of the growth measurement        |
| child_id (FK)             | Integer   | Child whose growth was measured            |
| measured_by (FK)          | Integer   | Staff member who recorded the measurements |
| measurement_date          | Date      | Date, measurements were taken              |
| age_in_days               | Integer   | Child's age in days at measurement         |
| age_in_months             | Integer   | Age in completed months                    |
| weight_kg                 | Decimal   | Measured weight in kilograms               |
| height_cm                 | Decimal   | Measured height in centimeters             |
| head_circumference_cm     | Decimal   | Measured head circumference in centimeters |
| muac_cm                   | Decimal   | Mid-upper arm circumference in centimeters |
| Bmi                       | Decimal   | Body mass index calculation                |
| weight_for_age_z_score    | Decimal   | Weight-for-age z-score                     |
| height_for_age_z_score    | Decimal   | Height-for-age z-score                     |
| weight_for_height_z_score | Decimal   | Weight-for-height score                    |

| Field         | Data Type     | Description                                           |
|---------------|---------------|-------------------------------------------------------|
| growth_status | Text          | Child's growth condition(underweight, overweight,...) |
| Remarks       | Text          | Notes about measurement                               |
| created_at    | Date and time | Date and time when the record was created             |

Table 4: Growth Measurement

## T16: Vaccine Master

Table Name: Vaccine Master

Description: Used to store the list of vaccines recommended by the National Immunization Service and their administration information

Primary Key: vaccine\_id

References: None

| Field           | Data Type     | Description                                    |
|-----------------|---------------|------------------------------------------------|
| vaccine_id (PK) | Integer       | Unique id of the vaccine                       |
| vaccine_name    | Text          | Formal vaccine name                            |
| Description     | Text          | Purpose and other description                  |
| Route           | Text          | Route of administration oral or IV             |
| created_at      | Date and time | Date and time when the record was created      |
| updated_at      | Date and time | Date and time when the record was last updated |

Table 5: Vaccine Master

## T17: Vaccine Schedule

Table Name: Vaccine Schedule

Description: Used to store vaccine doses and their recommended administration ages

Primary Key : schedule\_id

References : vaccine\_master(vaccine\_id)

| Field                  | Data Type     | Description                                             |
|------------------------|---------------|---------------------------------------------------------|
| schedule_id (PK)       | Integer       | Unique id of the scheduled dose.                        |
| vaccine_id (FK)        | Integer       | Vaccine to which the dose belongs                       |
| dose_number            | Integer       | id number of the dose                                   |
| recommended_age_months | Integer       | Recommended age for administration in months            |
| description            | Text          | Description of the dosage                               |
| created_at             | Date and time | Date and time when the record was created               |
| updated_at             | Date and time | Date and time when the record was most recently updated |

Table 6: Vaccine Schedule

### T18: Vaccination Record

Table Name: Vaccination Record

Description: Used to store vaccines administered to children

Primary Key: vaccination\_id

References: child(child\_id), vaccine\_schedule(schedule\_id), staff(administered\_by)

| Field                  | Data Type     | Description                                                |
|------------------------|---------------|------------------------------------------------------------|
| vaccination_id (PK)    | Integer       | Unique id of the record                                    |
| child_id (FK)          | Integer       | Child who received the vaccination                         |
| schedule_id (FK)       | Integer       | Scheduled vaccine dose of the record                       |
| administered_by (FK)   | Integer       | Staff member who administered the vaccine                  |
| vaccination_date       | Date          | Date of administration                                     |
| batch_number           | Text          | Batch number of the vaccine dose                           |
| next_due_date          | Date          | Date on which the next dose is due.                        |
| vaccination_status     | Text          | Whether the dose was completed, delayed, missed or omitted |
| adverse_event_reported | Boolean       | Indicates whether an adverse event was reported            |
| side_effects           | Text          | Observed or reported effects after vaccination             |
| remarks                | Text          | Additional follow-up notes                                 |
| created_at             | Date and time | Date and time when the record was created                  |
| updated_at             | Date and time | Date and time when the record was most recently updated    |

Table 7: Vaccination Record

### T19: Supplement Master

Table Name: Supplement Master

Description: Used to store the list of nutritional supplements, descriptions, and number of units available

Primary Key: supplement\_id

References: None

| Field              | Data Type     | Description                                             |
|--------------------|---------------|---------------------------------------------------------|
| supplement_id (PK) | Integer       | Unique id of the supplement                             |
| supplement_name    | Text          | Name of the product                                     |
| Description        | Text          | Supplement description                                  |
| Number of units    | Text          | Available amount                                        |
| created_at         | Date and time | Date and time when the record was created               |
| updated_at         | Date and time | Date and time when the record was most recently updated |

Table 20: Supplement Master

## T20: Supplement Distribution

Table Name: Supplement Distribution

Description: Stores supplements distributed to pregnant mothers, lactating mothers and children

Primary Key: distribution\_id

References: person(person\_id), supplement\_master(supplement\_id), staff(distributed\_by)

| Field             | Data Type      | Description                                             |
|-------------------|----------------|---------------------------------------------------------|
| distribution_id   | Integer        | Unique id of the distribution                           |
| person_id         | Integer        | Receiver of the supplement                              |
| supplement_id     | Integer        | Supplement product that was issued                      |
| distributed_by    | Integer        | Staff member who issued the supplement                  |
| distribution_date | Date           | Date the supplement was issued                          |
| quantity          | Decimal number | Amount issued in no. of units                           |
| expiry_date       | Date           | Expiry date of the issued stock                         |
| remarks           | Text           | Noted related to distribution                           |
| created_at        | Date and time  | Date and time when the record was created               |
| updated_at        | Date and time  | Date and time when the record was most recently updated |

Table 8: Supplement Distribution

## T21: Clinic Session

Table Name: Clinic Session

Description: Used to store planned clinic sessions

Primary Key: clinic\_session\_id

References: moh\_area(moh\_area\_id), staff(conducted\_by)

| Field                  | Data Type     | Description                                 |
|------------------------|---------------|---------------------------------------------|
| clinic_session_id (PK) | Integer       | Unique id of the clinic session             |
| moh_area_id (FK)       | Integer       | MOH area responsible for the clinic session |
| conducted_by (FK)      | Integer       | Staff member responsible for the session    |
| session_type           | Text          | Service types offered at the clinic         |
| clinic_date            | Date          | Date on which the clinic is held            |
| start_time             | Time          | Planned starting time                       |
| end_time               | Time          | Planned ending time                         |
| location               | Text          | Venue of the clinic session                 |
| remarks                | Text          | Notes for the session                       |
| created_at             | Date and time | Date and time when the record was created   |

| Field      | Data Type     | Description                                             |
|------------|---------------|---------------------------------------------------------|
| updated_at | Date and time | Date and time when the record was most recently updated |

Table 9: Clinic Session

## T22: Clinic Appointment

Table Name: Clinic Appointment

Description: Used to store appointments assigned to people for clinic sessions and track attendance

Primary Key: appointment\_id

References: clinic\_session(clinic\_session\_id), person(person\_id), pregnancy(pregnancy\_id)

| Field                  | Data Type     | Description                                             |
|------------------------|---------------|---------------------------------------------------------|
| appointment_id (PK)    | Integer       | Unique id of the appointment                            |
| clinic_session_id (FK) | Integer       | Clinic session to which the appointment belongs         |
| person_id (FK)         | Integer       | Person who attends the appointment                      |
| pregnancy_id (FK)      | Integer       | Related pregnancy id                                    |
| appointment_datetime   | Date and time | Scheduled date and time of attendance                   |
| appointment_type       | Text          | Purpose of the clinic appointment                       |
| appointment_status     | Text          | Current or completion state                             |
| attended               | Boolean       | Whether the person attended or not                      |
| attendance_time        | Date and time | Date and time of attendance                             |
| remarks                | Text          | Additional notes                                        |
| created_at             | Date and time | Date and time when the record was created               |
| updated_at             | Date and time | Date and time when the record was most recently updated |

Table 10: Clinic Appointment

## T23: Field Visit

Table Name: Field Visit

Description: Stores home visits performed for mothers or newborn babies

Primary Key: visit\_id

References: family(family\_id), staff(staff\_id), person(person\_id), pregnancy(pregnancy\_id)

| Field             | Data Type | Description                                      |
|-------------------|-----------|--------------------------------------------------|
| visit_id (PK)     | Integer   | Unique id of the field visit                     |
| family_id (FK)    | Integer   | Id of the family visited                         |
| staff_id (FK)     | Integer   | Staff member who did the visit                   |
| person_id (FK)    | Integer   | Person who the visit was for, if any             |
| pregnancy_id (FK) | Integer   | The pregnancy addressed during the visit, if any |

| Field              | Data Type     | Description                                               |
|--------------------|---------------|-----------------------------------------------------------|
| visit_date         | Date          | Date of the field visit                                   |
| visit_type         | Text          | Reason or type of visit                                   |
| visit_status       | Text          | Whether the visit was completed, cancelled or rescheduled |
| follow_up_required | Boolean       | Indicates whether another action is needed                |
| follow_up_date     | Date          | Date suggested for the next action or visit               |
| observations       | Text          | Findings and services recorded during the visit           |
| risk_identified    | Boolean       | Indicates whether a health risk was detected              |
| created_at         | Date and time | Date and time when the record was created                 |
| updated_at         | Date and time | Date and time when the record was last updated            |

Table 11: Field Visit

## T24: Report Submission

Table Name: Report Submission

Description: Used to store periodic report submissions and their approval status

Primary Key: report\_id

References: staff(submitted\_by), staff(reviewed\_by)

| Field                  | Data Type     | Description                                      |
|------------------------|---------------|--------------------------------------------------|
| report_id (PK)         | Integer       | Id of the submitted report                       |
| submitted_by (FK)      | Integer       | Staff member who submitted the report            |
| reviewed_by (FK)       | Integer       | Staff member who reviewed or approved the report |
| report_type            | Text          | Report category or type                          |
| reporting_period_start | Date          | First date covered by the report                 |
| reporting_period_end   | Date          | Last date covered by the report                  |
| submitted_date         | Date          | Report submission date                           |
| reviewed_date          | Date          | Report reviewed date                             |
| approval_status        | Text          | Current review decision                          |
| report_version         | Text          | Version number of the report                     |
| remarks                | Text          | Reviewer comments, or other notes                |
| created_at             | Date and time | Date and time when the record was created        |
| updated_at             | Date and time | Date and time when the record was last updated   |

Table 12: Report Submission

## Chapter 5. Implementation

### 5.1 Technologies Used

The selection of suitable technologies helped to develop the Public Health Management System as an effective web-based application. The main technologies used in this project are as follows:

**HTML:** HTML was used to create the basic structure of the web pages. It was used to design forms, tables, buttons, and other user interface components.

**CSS:** CSS was used to design and enhance the system's look. It helped in designing a user-friendly layout with suitable colors, styles and responsive designs.

**JavaScript:** JavaScript was used to add interactive features to the application. It was used for form validation, dynamic content updates, and improving user interaction.

**PHP:** PHP was used for server-side development. It handles user requests, processes data, manages system functions, and communicates with the database. Specifically object-oriented PHP was used rather than procedural PHP to minimize boilerplate code.

**MySQL:** MySQL was selected as the database management system for storing and managing system information. It stores data such as user details, pregnancy records, child records, growth measurements, development milestones, and vaccination records.

**Laragon:** Laragon was used as the local development server. It provides Apache and MySQL services that allow the system to run and be tested during development.

**TablePlus and DBeaver:** Both database management tools were used to manage the MySQL database. It helped create database tables, manage relationships, and maintain stored information.

These technologies provided the required support to develop a reliable, secure, and user-friendly Public Health Management System. The combination of frontend, backend, and database technologies helped to successfully implement the proposed solution.

### 5.2 System Architecture

The Midwifery Outcomes Management System was developed using a system architecture that follows View, Controller, Service, Repository data flowing format. This structure makes it possible to add new modules or modify existing functions without making changes throughout the entire application. The web architecture follows a client-server model where users access

the web application through a browser, and the application communicates with the central database server.

The frontend of the system was developed using HTML, CSS, and JavaScript to provide a simple and user-friendly interface. The design was created to support different screen sizes and provide a better experience for users.

In the backend, PHP handles the system operations, user requests, and communication between the application and the database. MySQL is used for storing and managing important healthcare information such as user details, pregnancy records, child information, growth records, development milestones, and vaccination details.

The application includes authentication and role-based access control as part of its implementation. Users are required to log in to access the functions available to their assigned role. The system also separates access to functions according to the responsibilities of PHMs, supervisory staff, and other users.

This architecture helps the Public Health Management System provide a reliable and efficient solution for managing maternal and child health records while supporting future growth of the system.

### **5.3 Project & Environment Setup**

The entire MOMS project was implemented as a single stack using PHP. The 8.3.0 version of PHP was used as it provides latest up to date features. For database management 8.4.23 version of MySQL was used. For database connectivity PHP Data Objects API was used rather than older mysqli API. This provides easier migrations to different relational database management systems in the future.

The Apache server 2.4.66 version was used locally as a server container. To make development easier Laragon Software was used which bundles all these services together.

For z-score calculation WHO Anthro package for R statistical programming language was used. The functionality of this package was exposed through a Rest API as a service via a server software also written in R language. The Growth module uses this API to calculate and record z-scores for each growth measurement.

The class autoloader function which is available in PHP 8.0.0, was used load classes to PHP whenever it requires them, rather than calling `include()`, `require()`, `require_once()` and etc.

## 5.4 Project Architectural Components

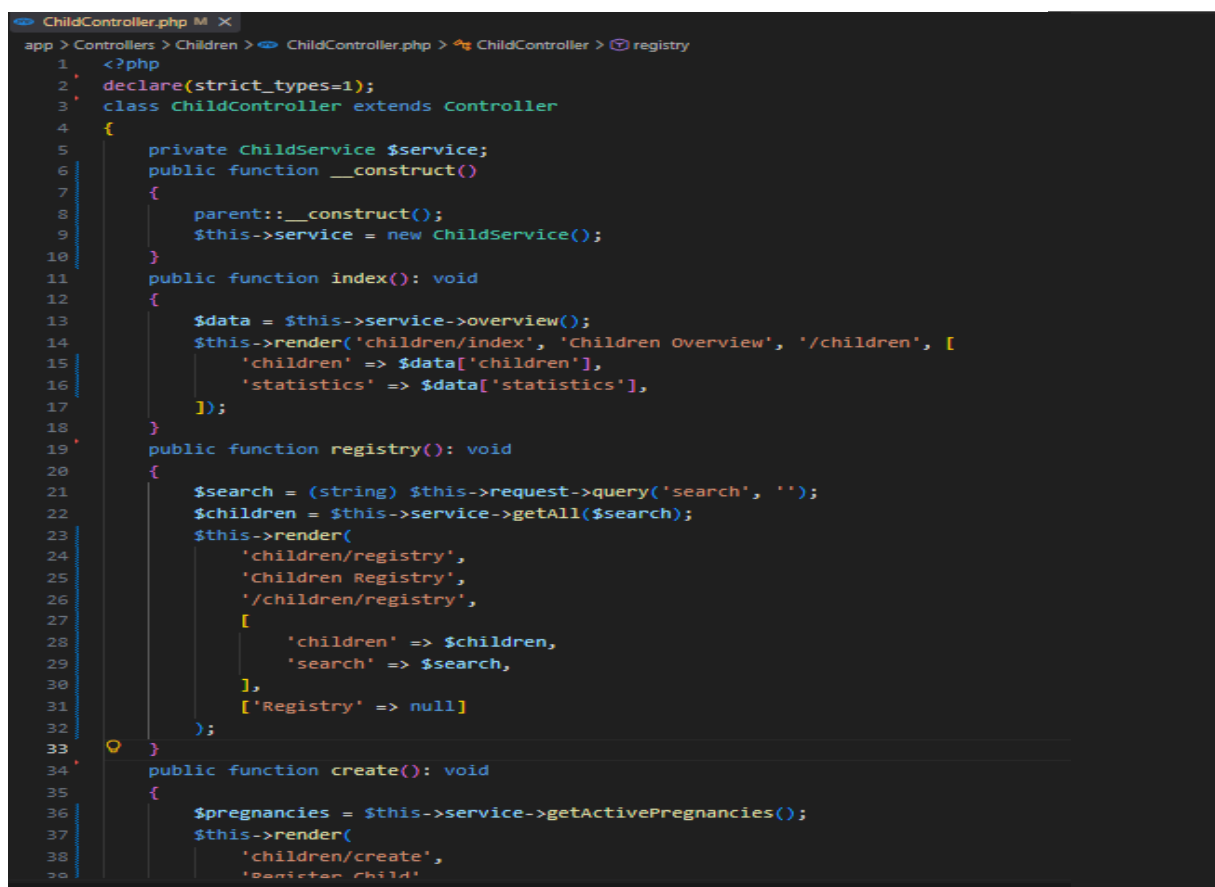
The Midwifery Outcomes Management System (MOMS) is implemented using a layered PHP application architecture. This architecture separates user interfaces, application logic, and database access into different components which are implemented by classes. This separation improves code organization, maintainability, and makes components easier to test.

The main source-code components of the MOMS application are:

- Routers
- Controllers
- Services
- Repositories
- Views
- Core application classes

### 1. Controllers

Controllers act as an intermediary between the user's request, the application logic, and the user interface. When a user performs an action such as registering a family, viewing a family profile, or creating a child record, the corresponding controller receives the request and determines what operation needs to be performed. And calls corresponding service layer method(s).



```
1 <?php
2 declare(strict_types=1);
3 class ChildController extends Controller
4 {
5     private ChildService $service;
6     public function __construct()
7     {
8         parent::__construct();
9         $this->service = new ChildService();
10    }
11    public function index(): void
12    {
13        $data = $this->service->overview();
14        $this->render('children/index', 'Children Overview', '/children', [
15            'children' => $data['children'],
16            'statistics' => $data['statistics'],
17        ]);
18    }
19    public function registry(): void
20    {
21        $search = (string) $this->request->query('search', '');
22        $children = $this->service->getAll($search);
23        $this->render(
24            'children/registry',
25            'Children Registry',
26            '/children/registry',
27            [
28                'children' => $children,
29                'search' => $search,
30            ],
31            ['Registry' => null]
32        );
33    }
34    public function create(): void
35    {
36        $pregnancies = $this->service->getActivePregnancies();
37        $this->render(
38            'children/create',
39            'Register Child'
40        );
41    }
42 }
```

Figure 44: Child Controller Class Implementation for Children Module

## 2. Services

The Service layer contains the application's business logic. A service therefore acts as the main layer responsible for implementing business rules and coordinating repositories.

```
app > Modules > Families > Services > FamilyService.php > FamilyService > getRegistry
1 <?php
2 declare(strict_types=1);
3 class FamilyService
4 {
5     private FamilyRepository $repository;
6     private FamilyMemberRepository $memberRepository;
7     private PHMAreaRepository $phmAreaRepository;
8     public function __construct()
9     {
10         $this->repository = new FamilyRepository();
11         $this->memberRepository = new FamilyMemberRepository();
12         $this->phmAreaRepository = new PHMAreaRepository();
13     }
14     // Get family registry with pagination
15     public function getRegistry(array $queryParams): array
16     {
17         // get page parameter from query parameters and validate
18         // integer and default to 1, minimum of 1
19         $page = filter_var(
20             $queryParams['page'] ?? 1,
21             FILTER_VALIDATE_INT,
22             [
23                 'options' => [
24                     'default' => 1,
25                     'min_range' => 1
26                 ]
27             ]
28         );
29         $recordsPerPage = 20;
30         $totalRecords = $this->repository->countRegistry();
31         $totalPages = max(1, (int) ceil($totalRecords / $recordsPerPage));
32         $page = min($page, $totalPages);
33         $offset = ($page - 1) * $recordsPerPage;
34         return [
35             'families' => $this->repository->findRegistry([], $recordsPerPage, $offset),
36             'pagination' => [
37                 'currentPage' => $page,
38                 'recordsPerPage' => $recordsPerPage,
39                 'totalRecords' => $totalRecords,
40                 'totalPages' => $totalPages,
41                 'from' => $totalRecords == 0 ? 0 : $offset + 1,
42                 'to' => min($offset + $recordsPerPage, $totalRecords),
43             ]
44         ];
45     }
46 }
47
```

Figure 45: Family Service Class Implementation for Family Module

### 3. Repositories

The Repository layer is responsible for communication with the database. Repositories contain database-related operations such as retrieving, inserting, updating, and deleting records. Each repository generally focuses on a particular database entity.

```
app > Modules > Children > Repositories > GrowthRepository.php > ...
4
5 class GrowthRepository extends Repository
6 {
7
8     // gets the latest growth measurements
9     // Used for: Recent measurement page - NI
10    // Dashboard "recent growth checks" widget
11    public function recent(int $limit = 100): array
12    {
13        return $this->findAll(
14            $this->selectSql() . ' ORDER BY gm.measurement_date DESC, gm.measurement_id DESC LIMIT :limit',
15            ['limit' => $limit]
16        );
17    }
18
19    // Gets all growth measurements of one child
20    // Used for: Child growth history chart.
21    public function findByChildId(int $childId): array
22    {
23        return $this->findAll(
24            $this->selectSql() . ' WHERE gm.child_id = :child_id
25            ORDER BY gm.measurement_date DESC, gm.measurement_id DESC',
26            ['child_id' => $childId]
27        );
28    }
29
30    // Gets one specific measurement record
31    // Used for: Edit/view a measurement
32    public function findRecord(int $childId, int $measurementId): ?array
33    {
34        return $this->findOne(
35            $this->selectSql() . ' WHERE gm.child_id = :child_id AND gm.measurement_id = :measurement_id LIMIT 1',
36            ['child_id' => $childId, 'measurement_id' => $measurementId]
37        );
38    }
39
40    // Finds children who are due for growth measurement
41    // Growth measurements requiring attention
42    public function dueChildren(): array
43    {
44        return $this->findAll(
45            'SELECT c.person_id AS child_id,
46            CONCAT_WS(" ", p.first_name, p.middle_name, p.last_name) AS full_name,
47            p.date_of_birth, f.registration_number AS family_code,
48            MAX(gm.measurement_date) AS last_measurement_date,
49            DATE_ADD(COALESCE(MAX(gm.measurement_date), p.date_of_birth), INTERVAL 1 MONTH) AS next_due_date,
50            CASE WHEN MAX(gm.measurement_date) IS NULL THEN "Never measured"
51            WHEN MAX(gm.measurement_date) < DATE_SUB(CURDATE(), INTERVAL 1 MONTH) THEN "Overdue"
52            ELSE "On track" END AS due_status'
53        );
54    }
55 }
```

Figure 46: Growth Repository Class Implementation for Growth Monitoring Module

## 4. Views

The View layer is responsible for presenting information to the user. Controller layer classes call view layer's `view()` method once the view data is available.

```
resources > views > dashboard > index.php > ...
1  <?php
2
3  $auth = new Auth();
4
5  $partialPath = ROOT_PATH . '/resources/views/partials';
6
7  ?>
8
9  <div class="space-y-6">
10
11     <!-- Stat cards -->
12     <?php require $partialPath . '/stat-cards.php'; ?>
13
14
15     <!-- =====
16     | Row 1
17     | Recent Families controls: scrollable target
18     | Attention Required controls: natural-height source
19     | ===== -->
20     <div
21         class="grid grid-cols-1 items-start gap-6 xl:grid-cols-3"
22         data-height-pair>
23         <!-- Match Attention Required height -->
24         <div
25             class="min-h-8 overflow-hidden xl:col-span-2"
26             data-height-target>
27             <?php require $partialPath . '/recent-activity.php'; ?>
28         </div>
29
30         <!-- Natural content height -->
31         <div
32             class="w-full self-start"
33             data-height-source>
34             <?php require $partialPath . '/attention-required.php'; ?>
35         </div>
36     </div>
37
38
39     <!-- =====
40     | Row 2
41     | Quick Actions controls: natural-height source
42     | Clinic Schedule: scrollable target
43     | ===== -->
44     <div
45         class="grid grid-cols-1 items-start gap-6 xl:grid-cols-2"
46         data-height-pair>
47         <!-- Natural content height -->
```

Figure 47: Index View of the Dashboard

## 5. Core Application Classes

These are common components that support the overall application such as components that handle URL routing, request handling, response handling, authentication and etc.

```
app > Core > Repository.php > Repository > query
1  <?php
2  declare(strict_types=1);
3  abstract class Repository
4  {
5      protected PDO $db;
6      // get database connection
7      public function __construct()
8      {
9          $this->db = (new Database())->connection();
10     }
11     //Prepare and execute any SQL statement
12     protected function query(
13         string $sql,
14         array $params = []
15     ): PDOStatement {
16         $statement = $this->db->prepare($sql);
17         $statement->execute($params);
18         return $statement;
19     }
20     // Fetch one row or return null
21     protected function findOne(
22         string $sql,
23         array $params = []
24     ): ?array {
25         $row = $this->query($sql, $params)->fetch();
26         return $row ?: null;
27     }
28     // Fetch multiple rows
29     protected function findAll(
30         string $sql,
31         array $params = []
32     ): array {
33         return $this
34             ->query($sql, $params)
35             ->fetchAll();
36     }
37     // execute any SQL statement
38     protected function execute(
39         string $sql,
40         array $params = []
41     ): int {
42         return $this
43             ->query($sql, $params)
44             ->rowCount();
45     }
46 }
```

Figure 48: Repository Class Implementation Which Encapsulates PDO Methods

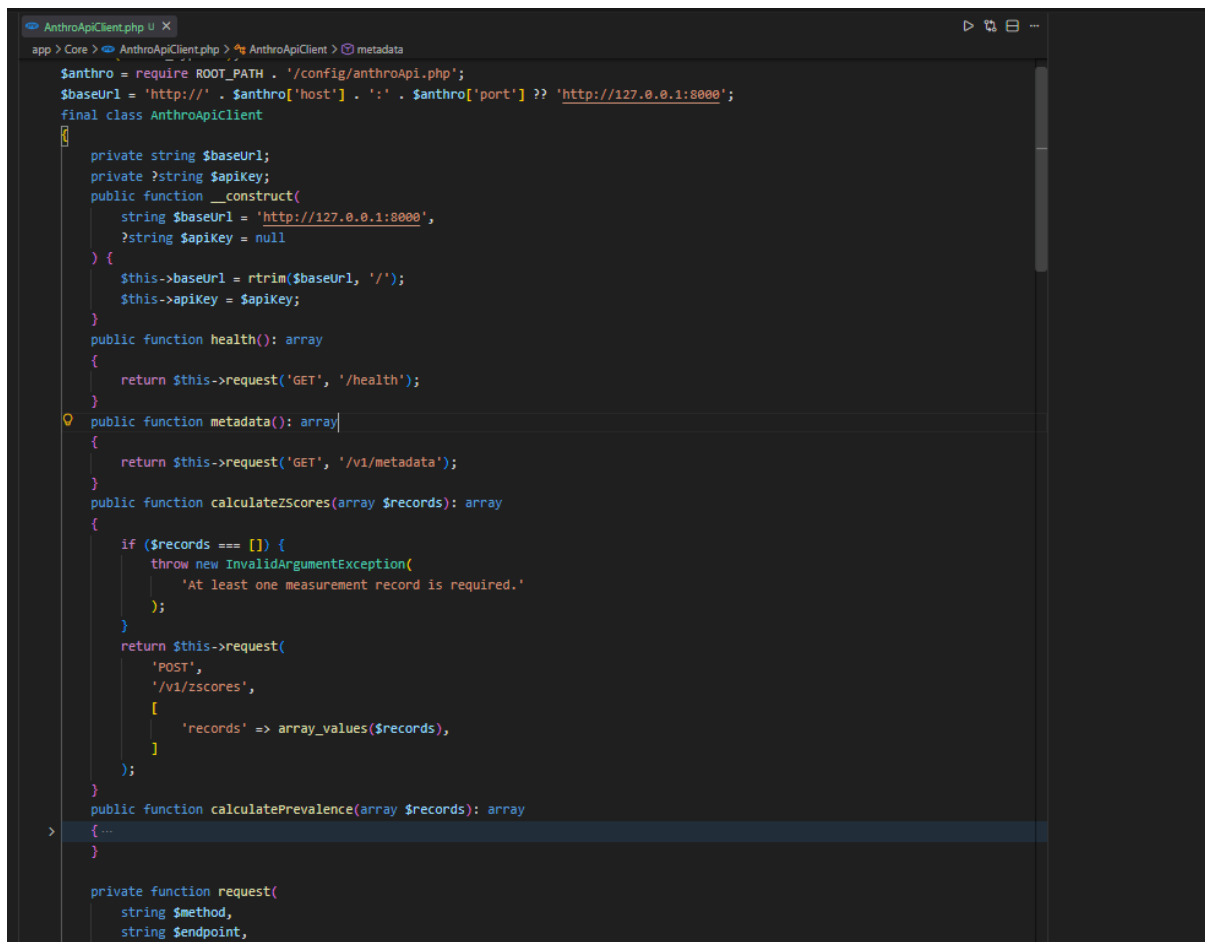
## 5.5 WHO Anthro Package for Child Growth Z-Score Calculation

The Anthro package is a library written in R, used in MOMS to calculate child growth z-scores based on the WHO child growth standards. As described in the solution design chapter, this package is integrated into the system via a separate REST API service, since it cannot be directly executed by the PHP web application. The PHP application sends the required child measurement data to the API, which then uses the R package to calculate and return the resulting z-scores to the application.

This section describes the integration of the Anthro package, including the PHP client class used to communicate with the API, and the API server that exposes the API endpoints for communication.

### 5.5.1 Anthro API Client Implementation

This PHP class is responsible for communicating with the Anthro REST API and handling the responses returned by the service.



```
AnthroApiClient.php U X
app > Core > AnthroApiClient.php > AnthroApiClient > metadata

$anthro = require ROOT_PATH . '/config/anthroApi.php';
$baseUrl = 'http://' . $anthro['host'] . ':' . $anthro['port'] ?? 'http://127.0.0.1:8000';
final class AnthroApiClient
{
    private string $baseUrl;
    private ?string $apiKey;
    public function __construct(
        string $baseUrl = 'http://127.0.0.1:8000',
        ?string $apiKey = null
    ) {
        $this->baseUrl = rtrim($baseUrl, '/');
        $this->apiKey = $apiKey;
    }
    public function health(): array
    {
        return $this->request('GET', '/health');
    }
    public function metadata(): array
    {
        return $this->request('GET', '/v1/metadata');
    }
    public function calculateZScores(array $records): array
    {
        if ($records === []) {
            throw new InvalidArgumentException(
                'At least one measurement record is required.'
            );
        }
        return $this->request(
            'POST',
            '/v1/zscores',
            [
                'records' => array_values($records),
            ]
        );
    }
    public function calculatePrevalence(array $records): array
    {
        ...
    }
    private function request(
        string $method,
        string $endpoint,
```

Figure 49: Anthro API Client Class Implementation

## 5.5.2 Using an Instance of the Anthro API Client

This client class is instantiated and used within the application service responsible for recording child growth measurements.

```
app > Modules > Children > Services > GrowthService.php > GrowthService > create
6  class GrowthService
7  {
8      private GrowthRepository $growthRepository;
9      private ChildRepository $childRepository;
10     private AnthroApiClient $anthroApi;
11
12     public function __construct()
13     {
14         $this->growthRepository = new GrowthRepository();
15         $this->childRepository = new ChildRepository();
16         $this->anthroApi = new AnthroApiClient('http://127.0.0.1:8000');
17     }
18     public function create(int $childId, array $data): ?int
19     {
20         // find whether this child exist
21         $child = $this->childRepository->findById($childId);
22         if ($child === null) {
23             return null;
24         }
25         // validate form data
26         if (!$this->validate($data)) return null;
27
28         // prepare the data for api call
29         $preparedData = $this->prepareMeasurementData($child, $data);
30
31         // format anthro analyzer request data
32         $anthroPostRecord = $this->createAnthroPostRecord($child, $preparedData);
33
34         try {
35             // api call to anthro analyzer
36             $response = $this->anthroApi->calculateZScores(
37                 [$anthroPostRecord]
38             );
39             // extract data from response body
40             if ($response['ok'] === false) {
41                 $_SESSION['errors'] = [
42                     'anthro' =>
43                         'The growth assessment service is unavailable. '
44                         . $response['error'],
45                 ];
46                 return null;
47             }
48             $result = $response['results'][0] ?? null;
49             $measurements = $result['measurements'] ?? null;
50             $interpretations = $result['interpretation'] ?? null;
51         }
```

Figure 50: Using Anthro API Client

### 5.5.3 Anthro REST API Server Implementation

The Anthro REST API server exposes Anthro package functionality through an HTTP endpoint, implemented using R and Plumber. The server receives child measurement data from the PHP application, passes the data to the Anthro package for calculation, and returns the calculated growth z-scores as a response.

```
D: > MOMS project_NIBM HDSE final > WHO child growth standards > Anthro API exposed > moms-anthro-api > plumber.R
258  /* Calculates WHO child-growth z-scores for one or more children.
259  /* @post /v1/zscores
260  /* @parser json
261  /* @serializer json
262  function(req, res) {
263    tryCatch({
264      body <- read_request_body(req)
265      vectors <- build_vectors(body$records)
266      scores <- anthro::anthro_zscores(
267        sex = vectors$sex,
268        age = vectors$age,
269        is_age_in_month = FALSE,
270        weight = vectors$weight,
271        lenhei = vectors$lenhei,
272        measure = vectors$measure,
273        headc = vectors$headc,
274        armc = vectors$armc,
275        triskin = vectors$triskin,
276        subskin = vectors$subskin,
277        oedema = vectors$oedema
278      )
279
280      list(
281        ok = TRUE,
282        package_version = as.character(utils::packageVersion("anthro")),
283        count = nrow(scores),
284        results = format_zscore_records(scores, vectors$id)
285      )
286    }, error = function(error) {
287      api_error(res, 400, conditionMessage(error))
288    })
  }
```

Figure 51: Anthro API Implementation

Here the “/v1/zscores” endpoint accepts the measurement data sent by the PHP application and uses the Anthro package to calculate the required growth indicators. The calculated results are parsed into JSON and returned to the PHP application as an API response.

## **5.6 Challenges and Lessons**

During the implementation of the Public Health Management System, several challenges were faced. These challenges helped to improve the understanding of software development and provided valuable experience.

### **1. Database Design and Management**

Designing the database structure was one of the main challenges during the development process. Creating proper relationships between tables such as users, parents, children, pregnancy records, growth records, and vaccination records required careful planning. A well-organized database was necessary to store and retrieve healthcare information correctly.

### **2. System Module Integration**

Connecting different modules of the system was another challenge. Modules such as child registration, growth monitoring, development milestone tracking, vaccination management, and report generation needed to work together properly. Testing and debugging were required to ensure smooth communication between different parts of the system.

### **3. Data Security**

Protecting sensitive healthcare information was an important challenge. Proper user authentication and access control were implemented to make sure that only authorized users could access specific information. Security measures were considered to protect personal and health-related data stored in the system.

### **4. Learning New Technologies**

Learning and applying technologies such as PHP Data Objects, class auto loading, Tailwind CSS, Chart.js, and web development techniques required additional effort. Understanding new concepts and using them correctly required research, testing, and continuous improvement.

## **5.7 Implementation Summary**

The development of the Midwifery Outcomes Management System followed a systematic approach to convert the system design into a fully functional web application. During the implementation process, attention was given to creating a system that is reliable, easy to use, and suitable for managing maternal and child health records.

The web application was developed using HTML, CSS, JavaScript, and PHP. These technologies were used to create the user interface, handle system operations, and manage

communication between the application and the database. The design was created with focus on simplicity and usability so that different users can easily access the required features.

MySQL was used as the database management system to store important healthcare information. Proper database design was implemented to maintain relationships between different records such as users, parents, children, pregnancy details, growth measurements, development milestones, and vaccination records.

Different modules of the system were integrated carefully to ensure smooth operation. User authentication, record management, and data processing functions were tested to confirm that the system works correctly.

During the implementation process, challenges related to database design, module integration, security, and learning new technologies were managed through proper planning and testing. These challenges helped improve the quality of the system and provided valuable experience in software development.

Overall, the implementation of the Public Health Management System was completed with focus on improving healthcare record management, reducing paperwork, and providing an efficient solution for MOH officers, Public Health Midwives, and parents.

## Chapter 6. Testing

### 6.1 Introduction

This chapter describes how product testing was carried out to ensure that the implemented MOMS system complies with the functional requirements of the proposed system. Also, some of the manual test cases are provided here to evaluate functionality of user login, family registration.

### 6.2 Test Plan

Functional tests were carried out to evaluate system's performance against the intended outputs. Each implemented feature was tested manually to ensure that the system behaved as expected. Any identified issue was corrected and retested before proceeding to the next feature implementation. Navigation elements and database operations were tested and corrected, until they behaved as expected.

All tests were conducted manually. Automated and programmatic tests were not performed because, the entire system was designed and implemented as a single stack. Therefore, it was not possible to perform separate backend testing. The other reason was that due to earlier system architectural decision implementation did not involve dependency injection. Therefore, it was difficult to test functional units programmatically and due to these two reasons programmatic tests were not implemented.

### 6.3 Sample Test Cases

| Test ID | Module | Test Case                                 | Description                                        | Expected Result                                             | Status |
|---------|--------|-------------------------------------------|----------------------------------------------------|-------------------------------------------------------------|--------|
| TC1     | Login  | Using all valid credentials               | Sign in user to the system                         | User is redirected to the dashboard                         | Pass   |
| TC2     | Login  | Using valid username but invalid password | User attempts to login with invalid credentials    | Login error is displayed "invalid username and/or password" | Pass   |
| TC3     | Login  | User try to logout                        | User attempts to sign out from the system          | User returned to the login interface                        | Pass   |
| TC4     | Family | View family registry                      | User tries to view the list of registered families | Relevant family records are displayed                       | Pass   |
| TC5     | Family | Register family with valid data           | User registers a new family in the system          | Family is successfully registered. Success message          | Pass   |

|     |        |                                                             |                                                                     |                                                                                                                                                                          |      |
|-----|--------|-------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|     |        |                                                             |                                                                     | displayed. User returned to newly created family record.                                                                                                                 |      |
| TC6 | Family | Register family with missing required data                  | All required family data for successful registration are not filled | Family is not registered. Failure message is displayed. User returned to the registration form with required fields highlighted and already inserted form data included. | Pass |
| TC7 | Family | View family details of existing family                      | User views details of an existing family                            | Correct family details are displayed                                                                                                                                     | Pass |
| TC8 | Family | View family details of a family not available in the system | Attempt to view details of non-existing family                      | Family not found message is displayed                                                                                                                                    | Pass |

Table 26: Sample Test Cases

## Chapter 7. Conclusion

The main purpose of this project was to develop a digital system to improve the management of maternal and child health records in the MOH office. The system was developed to replace the existing paper-based record system and provide a more efficient way to manage healthcare information.

The Midwifery Outcomes Management System provides important features such as user management, pregnancy registration, child registration, growth monitoring, development milestone tracking, vaccination management, and report generation. These features help Public Health Midwives, MOH officers, administrators, and parents manage health information more easily.

During the development of this project, technologies such as HTML, CSS, JavaScript, PHP, and MySQL were used to create the system. The database was designed to store health records securely, and different user roles were implemented to control access to information.

There were some challenges during the development process, such as designing the database structure, connecting different system modules, and ensuring that the system functions worked correctly. Solving these challenges helped to improve the developer's knowledge and understanding of software development.

Although the system achieved its main objectives, there are several improvements that can be added in the future. These improvements may include developing a mobile application, adding SMS or email notifications for clinic dates and vaccination reminders, improving report generation features, and adding more advanced health data analysis functions.

In conclusion, the Midwifery Outcomes Management System successfully provides a solution for improving maternal and child health record management. The system reduces paperwork, saves time, improves data accuracy, and helps healthcare workers provide better services to the community. This project shows how digital technology can support and improve healthcare management processes.

## References

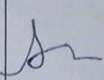
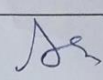
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## Appendix

### Supervisor Meet-up Log Sheet

#### Supervisor Meeting Schedule (Final Project)

Supervisor Name : Mrs. Shamali Kurukulasuriya

| <u>Date</u> | <u>Comment</u>                                                                 | <u>Signature</u>                                                                      |
|-------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 21/05/2026  | finalize the analysis →<br>ER and use case                                     |    |
| 22/06/2026  | finalize the ER diagram                                                        |    |
| 12/07/2026  | finalize the ER and<br>other diagram / analysis<br>continue the implementation |   |
| 27/07/2026  | complete the functionality<br>and Report                                       |  |
|             |                                                                                |                                                                                       |
|             |                                                                                |                                                                                       |

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